

Acute knee injuries

with RICHARD FROBELL, RANDALL COOPER, HAYDEN MORRIS and
MARK HUTCHINSON

The thing is I have no ACL. So unless I get surgery, there's nothing really magical that I can do that's going to make it better. I just can get my leg stronger, my muscle stronger and try and support it a little more. But that has a small impact. My knee is loose and it's not stable and that's the way it's going to be from here on out.

Lindsey Vonn, Olympic gold medal skier

For many athletes, the most fearful injury is that of the acute knee—it can spell the end of a professional career. Even for recreational athletes, an acute knee injury may be the catalyst for early arthritis. Acute knee injuries are common in all sports that require twisting movements and sudden changes of direction, especially the various types of football, basketball, netball and alpine skiing.

FUNCTIONAL ANATOMY

The knee joint can be divided into two parts: the tibio-femoral joint with its associated collateral ligaments, cruciate

ligaments and menisci; and the patellofemoral joint, which obtains stability from the medial and lateral retinaculum and the large extensor mechanism tendons (quadriceps and patella tendons) which encase the patella distally before its insertion on the proximal tibia. Most commonly we refer to the tibiofemoral joint as the knee joint. The anatomy of the knee joint is shown in Figure 35.1.

The two cruciate ('cross') ligaments, anterior and posterior, are often referred to as the 'crucial' ligaments, because of their importance in providing knee stability. They are named anterior and posterior in relation to their

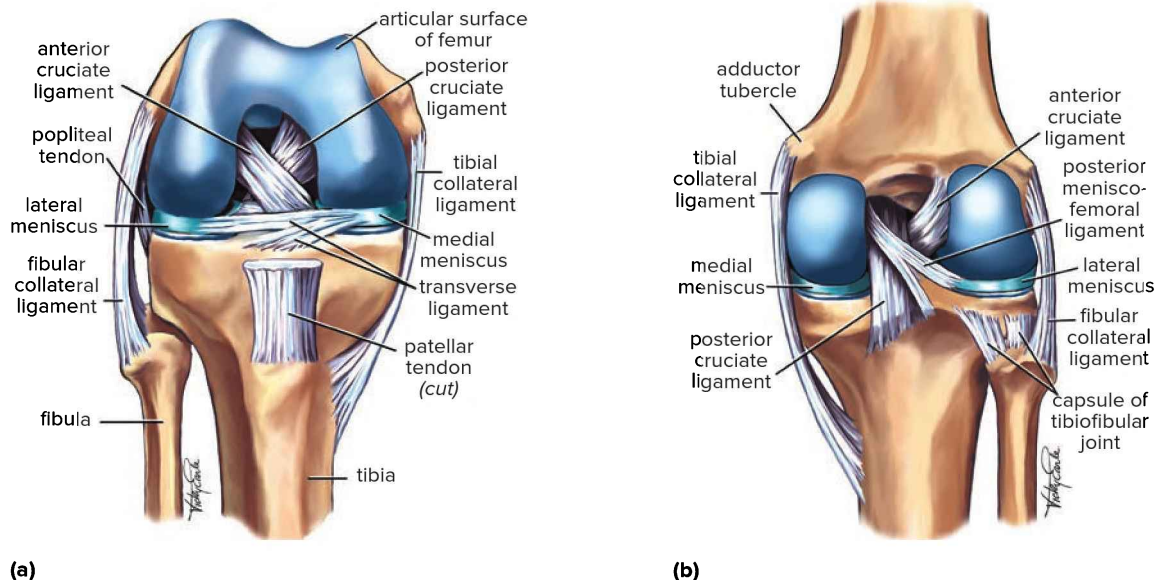


Figure 35.1 Anatomy of the knee joint (a) The knee joint (anterior view) (b) The knee joint (posterior view)

attachment to the tibia. The anterior cruciate ligament (ACL) prevents forward movement of the tibia in relation to the femur and controls rotational movement of the tibia under the femur. The posterior cruciate ligament (PCL) prevents the femur from sliding forwards off the tibial plateau.

The ACL is essential for control in pivoting movements. If the ACL is not functional, the tibia may rotate under the femur in an anterior-lateral direction such as when an athlete attempts to land from a jump, pivot or stop suddenly. The PCL stabilises the body (femur) above the tibia. In its absence, the femur wants to shift forwards on the tibia. This shift forwards is accentuated when one tries to run down an incline plane or down stairs.

The two collateral ligaments, the medial and lateral, provide medial and lateral stability to the knee joint. The superficial medial (or tibial) collateral ligament (MCL) is extracapsular. The deep layer, or coronal ligaments, attaches to the joint margins and has an attachment from its deep layer to the medial meniscus. The MCL prevents excessive medial opening (i.e. valgus) of the tibial-femoral joint.

The lateral (or fibular) collateral ligament (LCL) is a narrow strong cord with no attachment to the lateral meniscus. It prevents lateral opening of the tibia on the femur during varus stress; it also stabilises the upright knee in single-leg stance phase of gait.

The medial and lateral menisci are intra-articular and attach to the capsule layer at the level of the joint line. The menisci buffer some of the forces placed through the knee joint and protect the otherwise exposed articular surfaces from damage. By increasing the concavity of the tibia, they contribute to stabilising the knee. As the menisci contribute to joint lubrication and nutrition, it is important to preserve as much of them as possible after injury.

CLINICAL APPROACH

The acute knee injury of greatest concern to the athlete is the tear of the ACL. Meniscal injuries are common among

athletes, either in isolation or combined with a ligament injury (e.g. of the MCL or ACL). Importantly, the articular cartilage of the knee is often damaged in association with ligament or meniscal injuries. Cartilage damage, depending on the size and/or location, can accelerate the development of arthritis.¹

A list of acute knee injuries occurring in sport is shown in Table 35.1.

‘Does this patient have a significant knee injury?’

To answer this critical question, consider the following elements of the patient’s story:

- the mechanism of injury (low-energy force versus high-energy force)
- the amount of pain and disability at the time of injury
- the presence and timing of onset of swelling which may be a clue to haemarthrosis (joint bleeding)
- the degree of disability on presentation to the clinician
- patient-specific vulnerabilities (previous injury, medical comorbidities that can affect bone and/or tendon health).

In the majority of cases, an acute knee injury can be diagnosed with an appropriate history and examination. The main goals of assessment are:

1. to determine which structures have been damaged
2. to determine the extent of damage to each structure
3. to determine the degree of joint/limb disability to provide safe and timely initial management.

History

It is absolutely critical to invite the patient to tell his or her own story of the injury. Once the patient has explained what happened, the clinician may elicit additional aspects of the history.

Table 35.1 Causes of acute knee pain

Common	Less common	Not to be missed
Medial meniscus tear	Patellar tendon rupture	Fracture of the tibial plateau
MCL sprain	Quadriceps tendon rupture	Avulsion fracture of tibial spine
ACL sprain (rupture)	Acute patellofemoral contusion	Osteochondritis dissecans (in adolescents)
Lateral meniscus tear	LCL sprain	Complex regional pain syndrome
Articular cartilage injury	Bursal haematoma/bursitis	type 1 (post injury)
PCL sprain	Acute fat pad impingement	Quadriceps muscle rupture
Patellar dislocation	Avulsion of biceps femoris tendon	
	Superior tibiofibular joint injury	

Note: all these conditions may occur in isolation or, commonly, in association with other conditions

Important components include:

- a description of the precise mechanism of injury and the subsequent symptoms (e.g. pain and giving way)
- demonstration by the patient if possible, on the uninjured knee, of the stress applied at the time of injury
- the location of pain – pain associated with cruciate ligament injuries is often poorly localised (or emanates from the lateral tibial plateau); pain from injuries to the collateral ligaments is usually fairly well localised
- severity of pain – this does not always correlate with the severity of the injury, although most ACL injuries are usually painful immediately.

PRACTICE PEARL

The degree and time of onset of swelling can reflect either intra-articular or extra-articular injury and thus, provides an important clue to the injured structure (Table 35.2). Intra-articular swelling (i.e. haemarthrosis) is usually voluminous (obvious) and develops within 1 or 2 hours following injury.

The causes of haemarthrosis are:

- major ligament rupture
 - ACL
 - PCL
- patellar dislocation
- osteochondral fracture
- peripheral tear of the meniscus, more common medially
- Hoffa's syndrome (acute fat pad impingement)
- bleeding diathesis (rare).

Note: Lipohaemarthrosis (fat and blood in the knee) is caused by intra-articular fractures. Lipohaemarthrosis will present in a similar manner to haemarthrosis.

PRACTICE PEARL

An effusion that develops after a few hours or, more commonly, the following day represents reactive synovitis and is a feature of meniscal and chondral injuries. There is usually little effusion with collateral ligament injuries.

PRACTICE PEARL

If the patient volunteers having heard a 'pop', a 'snap' or felt a 'tear', the injury should be considered as an ACL tear until proven otherwise.

Patients presenting with a sensation of something having 'moved' or 'popped out' in the knee may be thought to have a patellar dislocation. However, this symptom is more commonly associated with an ACL rupture. There may be associated 'clicking' or 'locking' and this is often seen with meniscal injuries. Locking is classically associated with a loose body or displaced meniscal tear. Locking does not mean locked in one knee position, but is used when significant loss of passive range of motion is present, especially loss of full extension. It is helpful to ask the patient in what 'position' the knee locks. If the patient reports that the knee locks when it is straight and does not bend, this usually is a manifestation of patellofemoral pain and injury – the kneecap is unable to engage in the groove secondary to pain.

The symptom of 'giving way' can occur with instability, such as in ACL deficiency. Instability may also occur with meniscal tears, articular cartilage damage, patellofemoral pain (Chapter 36) or severe knee pain. In the latter cases of knee *pain* instability, a careful history will reveal more of a 'jackknife' (collapsing) phenomenon in flexion rather than a true 'give way' in extension. Patients with recurrent patellar dislocation and those with loose bodies in the knee can describe similar sensations. If a patient reports feeling unstable on steps, this is most often a reflection of quadriceps weakness and/or pain and rarely represents true kneecap instability.

The comprehensive history will also include:

- the initial management of the injury
- the degree of disability
- a history of previous injury to either knee or any previous surgery
- the patient's age, occupation, type of sport and leisure activities and the level of sport played.

If the patient is a good historian, the diagnosis will be obvious in most cases.

Table 35.2 Time relationship of swelling to diagnosis

Intermediate (0–2 hours) – haemarthrosis	Delayed (6–24 hours) – effusion	No swelling
ACL rupture Patellar dislocation Major chondral lesion MCL sprain (deep)	Meniscus Smaller chondral lesion	MCL sprain (superficial)

Physical examination

Examination includes:

1. Observation
 - a. standing
 - b. walking
 - c. supine (Fig. 35.2a)
2. Active movements
 - a. flexion
 - b. extension
 - c. straight leg raise
3. Passive movements
 - a. flexion (Fig. 35.2b)
 - b. extension (Fig. 35.2c)
4. Palpation
 - a. patellofemoral joint (including patellar and quadriceps tendons, medial and lateral retinaculum)
 - b. MCL
 - c. LCL
 - d. medial joint line (Fig. 35.2d)
 - e. lateral joint line
 - f. posterior structures (e.g. hamstring tendons, Baker's cyst, gastrocnemius origins, best done in the prone position)

5. Special tests

- a. presence of effusion (Fig. 35.2e)
- b. stability tests
 - i. MCL (Fig. 35.2f)
 - ii. LCL (Fig. 35.2g)
 - iii. ACL
 1. Lachman's test (Figs 35.2h-k)
 2. anterior drawer test (Fig. 35.2l)
 3. pivot shift test (Fig. 35.2m)
- iv. PCL
 1. posterior sag (Fig. 35.2n)
 2. reverse Lachman's test
 3. posterior drawer test (Fig. 35.2o)
 4. external rotation test—active and passive
- v. patella
 1. medial and lateral patella translation (or mobility)
- c. flexion/rotation
 - i. McMurray's test (Fig. 35.2p)
 - ii. Apley's grind test
- d. patellar apprehension test (Fig. 35.2q)



(a)



(b)

Figure 35.2 Examination of the patient with an acute knee injury (a) Observation—supine. Look for swelling, deformity and bruising (b) Passive movement—flexion. Assess range of motion, end feel and presence of pain

- e. patellofemoral joint (Chapter 36)
- f. functional tests
 - i. squat test (helps to assess functional valgus collapse of knee)
 - ii. hopping on the spot
 - iii. pelvic bridge/plank arrow (helps assess core strength)

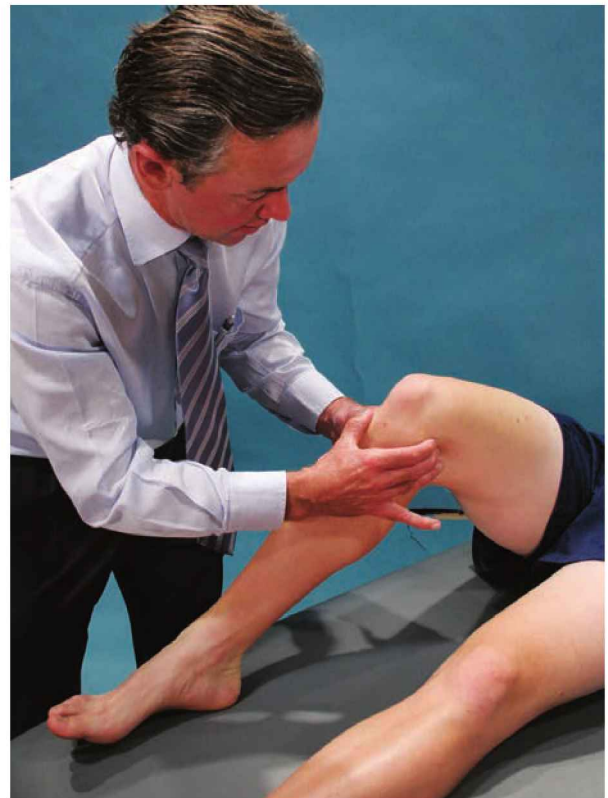
The clinical utility of the various special tests for diagnosing acute knee injuries is shown in Table 35.3.

Key outcome measures

Patient-reported outcome measures (PROMs) for acute knee injuries are shown in Table 35.4.



(c)



(d)



(e)

Figure 35.2 (cont.) (c) Passive movement—extension. Hold both legs by the toes, looking for fixed flexion deformity or hyperextension in ACL or PCL rupture. Overpressure may be applied to assess end range. This procedure may provoke pain in meniscal injuries (d) Palpation—medial joint line. The knee should be palpated in 45–90° of flexion (e) Special tests—presence of effusion. Manually drain the medial subpatellar pouch by stroking the fluid in a superior direction. Then ‘milk’ the fluid back into the knee from above on the lateral side while observing the pouch for evidence that fluid is re-accumulating. This test is more sensitive than the ‘patellar tap’. It is important to differentiate between an intra-articular effusion and an extra-articular haemorrhagic bursitis



(f)



(g)

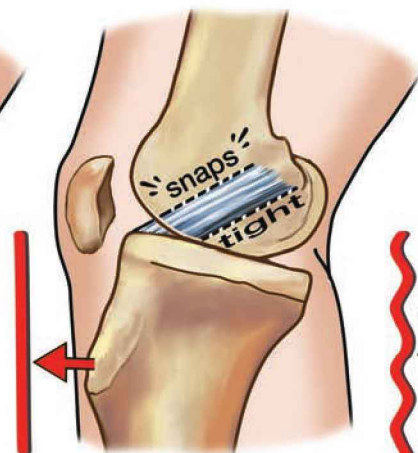


(h)

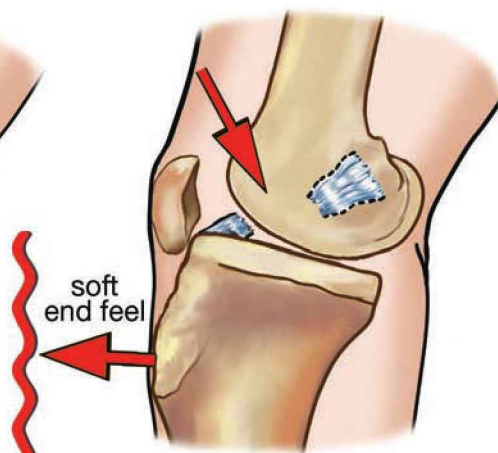
Figure 35.2 (cont.) (f) Stability test—MCL. This is tested first with the knee in full extension and then also at 30° of flexion (illustrated). The examiner applies a valgus force, being careful to eliminate any femoral rotation. Assess for onset of any pain, extent of valgus movement and feel for end point. If the knee 'gaps' at full extension, there must be associated posterior cruciate injury (g) Stability test—LCL. The LCL is tested in a similar manner to the MCL except with varus stress applied (h) Stability test—Lachman's test. Lachman's test is performed with the knee in 15° of flexion, ensuring the hamstrings are relaxed. The examiner draws the tibia forwards, feeling for laxity and assessing the quality of the end point. Compare with the uninjured side (i) The ACL is slightly slack in the start position (j) When the ACL is intact, the ligament snaps tight and the examiner senses a 'firm'/'sudden' end feel (k) When the ACL is ruptured, the Lachman's test results in a 'softer'/'gradual' end feel



(i)



(j)



(k)



(l)



(m)



(n)



(o)

Figure 35.2 (cont.) (l) Stability test—anterior drawer test. This is performed with the knee in 90° of flexion and the patient's foot kept stable. Ensure the hamstrings are relaxed with the index finger on the femoral condyles. The tibia is drawn anteriorly and assessed for degree of movement and quality of end point. The test can be performed with the tibia in internal and external rotation to assess anterolateral and anteromedial instability respectively (m) Special test—pivot shift test. With the tibia internally rotated and the knee in full extension, a valgus force is applied to the knee. In a knee with ACL deficiency, the condyles will be subluxated. The knee is then flexed, looking for a 'clunk' of reduction, which renders the pivot shift test positive. Maintaining this position, the knee is extended, looking for a click into subluxation, which is called a 'positive jerk test' (n) Stability test—posterior sag. With both knees flexed at 90° and the patient relaxed, the position of the tibia relative to the femur is observed. This will be relatively posterior in the knee with PCL deficiency (o) Stability test—posterior drawer test. With the knee as for the anterior drawer test, the examiner grips the tibia firmly as shown and pushes it posteriorly. Feel for the extent of the posterior movement and quality of end point. The test can be repeated with the tibia in external rotation to assess posterolateral capsular integrity



(p)



(q)

Figure 35.2 (cont.) (p) Flexion/rotation test—McMurray's test. The knee is flexed and, at various stages of flexion, internal and external rotation of the tibia are performed. The presence of pain and a palpable 'clunk' is a positive McMurray's test and is consistent with meniscal injury. If there is no clunk but the patient's pain is reproduced, then the meniscus may be damaged or there may be a patellofemoral joint abnormality (q) Special test—patellar apprehension test. The knee may be placed on a pillow to maintain 20–30° of flexion. Gently push the patella laterally. The test is positive if the patient develops apprehension with a sensation of impending dislocation

Table 35.3 Diagnostic accuracy of tests used in the acute knee injury setting

	What is a positive finding?	Sensitivity	Specificity	LR+	LR–
Knee fracture					
Ottawa knee decision rule ²	One of the following: • age $\geq$ 55 years • tenderness head of fibula • isolated tenderness of patella • inability to flex beyond 90° • inability to weight-bear more than 4 steps	98.5	49	1.9*	0.05
ACL injury					
Lachman's test ³	Soft end feel for tibial translation	81	81	4.5	0.22
Anterior drawer ³	>5 mm anterior tibial translation	38	81	4.5	0.67
Pivot shift ³	Anterior subluxation of tibia	28	81	5.4	0.84

Table 35.3 Cont.

	What is a positive finding?	Sensitivity	Specificity	LR+	LR–
PCL injury					
Posterior drawer ⁴	>5 mm posterior tibial translation	22–100	98	50.1	0.11
Posterior sag sign ⁴	Posterior sagging of tibia	46–100	100	88.4	0.28
Reverse Lachman's ⁴	Soft end feel for tibial translation	63	89	5.9	0.41
Meniscal tear					
McMurray's test ⁵	Palpable or audible click or pain	61	84	3.2	0.52
Joint line tenderness ⁵	Reproduction of patient's pain	83	83	4.0	0.52
Apley's test ⁶	Pain	61	70	2.0	0.56
MCL injury					
Valgus stress test at 30° ⁷	Pain	78	67	2.3	0.30
	Laxity	91	49	1.8	0.20
LCL injury					
Varus stress test at 30°	Pain/laxity	NA	NA	NA	NA

LR+ = positive likelihood ratio; LR– = negative likelihood ratio; NA = not available; see Chapter 14 for a guide to interpreting test sensitivity, specificity and likelihood ratios

* estimated from available research data

Table 35.4 Patient-reported outcome measures for acute knee injuries

International Knee Documentation Committee (IKDC) Subjective Knee Evaluation Form

- Self-administered questionnaire evaluating symptoms, function and sports activities due to knee impairment
- Eighteen items (7 items for symptoms, 10 items for sport activities and 1 item for current knee function)
- Score range 0–100
- Test-retest reliability: 0.845–0.93^{8–10} (ACL injuries) 0.95¹¹ (meniscal injuries)
- Construct validity: Lysholm score 0.62–0.89,^{10, 12–15} SF-36 physical function 0.71,¹⁶ Cincinnati Knee Rating System 0.87–0.946¹⁷
- MDC 95%: 8.8–15.6^{11, 18}
- MCID: (Sn) 11.5,¹⁹ (Sp) 20.5¹⁹
- Available online in a range of languages: American Orthopaedic Society for Sports Medicine, www.sportsmed.org/aossmimis

Knee Injury and Osteoarthritis Outcome Score (KOOS)

- Self-administered questionnaire evaluating a patient's opinion about their knee and associated problems²⁰
- Five domains: Symptoms and stiffness (7 items), Pain (9 items), ADL function (17 items), Sport and recreational activity function (5 items) and Quality of life (4 items)
- Score range 0–100 (0 = extreme symptoms and 100 = no symptoms)
- Test-retest reliability: Symptoms and stiffness (cartilage lesions 0.9/ACL 0.89), Pain (cartilage lesions 0.91/ACL 0.87), ADL function (cartilage lesions 0.93/ACL 0.87), Sport and recreational activity function (cartilage lesions 0.92/ACL 0.84) and Quality of life (cartilage lesions 0.95/ACL 0.88)²¹
- Construct validity: SF-36-bodily pain (cartilage lesions 0.72/ACL 0.66), SF-36 physical function (cartilage lesions 0.63/ACL 0.65)²¹
- MDC 95%: Symptoms and stiffness (cartilage lesions 30.8/ACL 15.6), Pain (cartilage lesions 26.3/ACL 12.7), ADL function (cartilage lesions 29.7/ACL 12.9), Sport and recreational activity function (cartilage lesions 29.9/19.2) and Quality of life (cartilage lesions 20.5/ACL 16.3)²¹
- MCID: NA
- Available online in a range of languages: Professor Ewa Roos, Institute of Sports Science and Clinical Biomechanics, www.koos.nu/

continued

Table 35.4 Cont.

Lysholm Knee Scoring Scale

- An eight-item knee-specific scale initially designed for the purpose of assessing knee ligament function that can be self-administered or clinician administered
- Eight items related to knee function
- Score range 0–100 (100 = no symptoms or disability)
- Test–retest reliability: 0.88–0.97^{15, 22–25}
- Construct validity: Fulkerson Knee Instability Scale 0.93,²⁵ Kujala Anterior Knee Pain Scale 0.86,²⁶ SF-12 physical function 0.54²⁴ and SF-12 role physical 0.48²⁴
- MDC95%: 8.9–10.1^{15, 22–25}
- MCID: Has not been established²⁰
- Available online in a range of languages; original scale is freely available from the paper:²² www.ncbi.nlm.nih.gov/pubmed/4028566

Lysholm Knee Scoring Scale

- A standardised method to grade work and sporting activity level; the questionnaire can be self-administered or clinician administered and was developed initially to be used in combination with the Lysholm knee score
- One item
- Score range 0–16 (16 = increased frequency of participation in athletic activity)
- Test–retest reliability 0.82–0.92^{15, 23, 25}
- Construct validity: SF-12 physical function 0.46,²³ Lysholm Knee Scoring Scale 0.35²⁴
- MDC95%: 1²³
- MCID: Has not been established²⁰
- Original scale is freely available from the original paper:²² www.ncbi.nlm.nih.gov/pubmed/4028566

Lysholm Knee Scoring Scale

- Self-administered, knee-specific questionnaire that evaluates activity level of various knee disorders²⁰
- Four items related to frequency of athletic activity²⁰
- Score range 0–16 (16 = more frequent participation in athletic activities)²⁰
- Test–retest reliability 0.97 (mixed knee pathologies)²⁷
- Construct validity: Tegner Activity Scale 0.66,²⁷ Cincinnati Knee Rating System 0.67²⁷
- MDC95%: Has not been established²⁰
- MCID: Has not been established²⁰
- Available online: www.aaos.org/uploadedFiles/PreProduction/Quality/Measures/pdf-MARX%20SCALE-%20english.pdf

SF = Short Form Health Survey; MDC = minimal detectable change; MCID = minimal clinically important difference;

Sn = sensitivity; Sp = specificity; ADL = activities of daily living

Investigations

Plain radiography (AP)

Should you perform a plain radiograph in cases of an acute knee injury? More than 90% of radiographs ordered to evaluate knee injuries are normal. The Ottawa knee X-ray decision rules may be helpful (Table 35.5).²⁸

The main aim of performing a plain radiograph in cases of moderate and severe acute knee injuries is to detect a fracture:

- tibial avulsion fracture associated with an ACL or PCL injury
- a tibial plateau fracture following a high-speed injury or knee loading injury
- an osteochondral fracture after patellar dislocation.

The classic radiograph series should include an anterior-posterior (AP) view, a lateral view and a patellar tangential view (skyline or sunrise view). Additional oblique views can assist in identifying subtle loose bodies. A notch view can image cruciate avulsions and assess notch width as a factor for ACL injury. Weight-bearing views are ideal to assess for associated arthrosis. For children, comparative images of the contralateral side may be helpful in identifying subtle physeal injuries. Finally, stress radiographs can assist in defining the severity of ligamentous injuries.

If surgery is considered, preoperative films are justified.

Magnetic resonance imaging

Magnetic resonance imaging (MRI) is reliable, safe and accurate in the diagnostic work-up of acute knee injuries.

Table 35.5 Criteria for knee X-ray based on Ottawa knee rule*

A knee radiograph is indicated after trauma only when at least one of the following is present:

- patient age more than 55
- tenderness at the fibular head
- tenderness over the patella
- inability to flex the knee to 90° (this captures most haemarthrosis, fractures)
- inability to weight-bear for four steps at the time of the injury and when examined.

To these, we suggest adding a high Index of suspicion for:

- high-speed injuries
- children or adolescents (who may avulse a bony fragment instead of tearing a cruciate ligament)
- if there is clinical suspicion of loose bodies.

* The Ottawa knee rule was designed for use in the emergency department setting

Even when a ligament disruption is diagnosed by physical exam, MRI can add value by demonstrating the extent of associated injury to cartilage and meniscus. Patellar and quadriceps tendon injury can also be assessed.

PRACTICE PEARL

MRI does not substitute for a good physical examination.

MRI should never be ordered in the absence of a thorough history and physical examination and MRI findings should be interpreted in the light of the clinical findings. For example, a relatively high proportion of asymptomatic knees, especially in athletes >35–40 years, have meniscus tears.²⁹ Thus, evidence of a meniscal tear on MRI, but not associated with pain, local tenderness and positive examination findings, might be a 'red herring' (irrelevant) not in need of treatment.

MRI can be very useful when a primary patellar dislocation is suspected clinically and can help to detect osteochondral avulsion fractures that require surgical assessment. As the avulsed fragment swells (and deteriorates), there is limited window of time for surgical fixation. Usually it needs to be identified and treated within 10–14 days of the injury.

Significant knee injuries are associated with MRI findings of oedema in the subchondral region—a bone bruise or bone marrow lesion (BML). Clinically, a BML is associated with pain, tenderness, swelling and delayed recovery. The presence of a BML indicates articular cartilage damage,^{30, 31} but its clinical relevance is discussed later in this chapter.

The role for diagnostic arthroscopy

The concept of 'diagnostic arthroscopy' predates knee MRI. Today, it remains a diagnostic option in settings where MRI is not available. In the developed world, arthroscopy may be performed to seek pertinent

intra-articular pathology in those rare cases when the patient has persistent pain and swelling not responding to treatment and MRI is normal.

Ultrasound examination

As outlined in Chapter 15, ultrasound examination can detect complete and partial patellar or quadriceps tendon tears, the size and location of bursal swelling and identify intra-versus extra-articular swelling if necessary. Ultrasound is not as efficacious for intra-articular injuries such as cruciate ligament or osteochondral injury. When radiography and MRI are available, ultrasound in a clinic/hospital setting adds little value to the assessment of the acute knee. As a sideline instrument, it can help confirm the diagnosis of a complete patellar or quadriceps tendon rupture and facilitate definite transfer, referral and treatment of that patient.

MENISCAL INJURIES

Acute meniscal tears occur when the shear stress generated within the knee in flexion and compression combined with femoral rotation exceeds the meniscal collagen's ability to resist these forces.³² The medial meniscal attachment to the medial joint capsule decreases its mobility, thereby increasing its risk for injury compared with the more mobile lateral meniscus.³³

Experienced clinicians consider patients with meniscal injury to be on a continuum where the two anchors of the continuum can be considered as being:

1. the young patient with the morphologically pristine articular surfaces in whom substantial forces damage the previously pristine meniscus, and
2. the older (>40 years) patient in whom an 'acute' presentation of knee pain arises from a knee that already has features of (asymptomatic) articular cartilage damage and has menisci that already had (asymptomatic) degenerative features.

Patients will fall along this continuum—and where they fall should influence management (see the section on treatment). Degenerative meniscal tears occur in the older population frequently without an inciting event and also without symptoms.²⁹ (See box below.)

The different types of meniscal tear are shown in Figure 35.3.

Clinical features

The history can provide a mechanism and a sense of the severity of meniscal tears. The clinical features can include the following.

- The most common mechanism of meniscal injury is a twisting injury with the foot anchored on the ground; this rotational force is often caused by another player's body.
- The twisting component may be of comparatively slow speed. This type of injury is commonly seen in football and basketball players.
- The degree of pain associated with an acute meniscal injury varies considerably. Some patients may describe a tearing sensation at the time of injury.
- A small meniscal tear may cause no immediate symptoms; it may become painful and cause knee swelling over 24 hours.
- Small tears may also occur with minimal trauma in the older athlete as a result of degenerative change of the meniscus.
- Patients with more severe meniscal injuries, such as a longitudinal ('bucket handle') tear, present with more

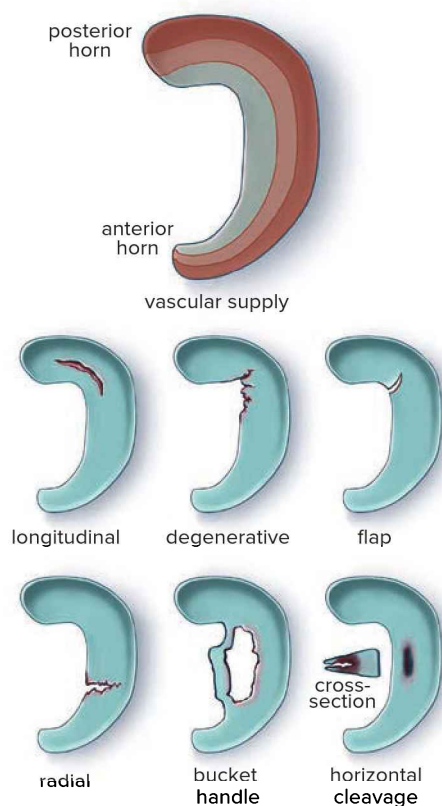


Figure 35.3 Meniscus tear orientation and zones of vascularity; these drawings are of a medial meniscal tear

Degenerative meniscus lesions

Adapted from the ESSKA Meniscus Consensus project (www.esska.org)

A degenerative meniscus lesion is a slowly developing lesion, typically involving a horizontal cleavage of the meniscus, generally in a middle-aged or older person. Such meniscus lesions are frequent in the general population and often incidental findings on knee MRI. There is often no clear history of an acute knee injury.

The prevalence of meniscus lesions in the general population:

- age 50–59 years ≈ 25%
- 60–69 years ≈ 35%
- 70–79 years ≈ 45%
- patients with knee osteoarthritis ≈ 75–95%.

There is very limited evidence that pain in the degenerative knee is directly attributable to a degenerative meniscus lesion, even if the lesion is considered to be unstable. Great caution must be taken before arriving at the conclusion that the degenerative meniscus lesion is the direct cause of the patient's knee symptoms.

Knee radiographs should be used as a first-line imaging tool to support a diagnosis of osteoarthritis or to detect certain more rare pathologies of the knee. Therefore, at least anteroposterior weight-bearing semi-flexed knee radiographs including a lateral view should be included in the work-up of the middle-aged or older patient with knee pain.

Knee MRI is typically not indicated in the first-line work-up of middle-aged or older patients with knee joint symptoms. However, knee MRI may be indicated in selected patients with refractory symptoms or in the presence of 'warning flags' or localised symptoms indicating more rare disease. If a surgical indication is considered, based on history, symptoms, clinical

exam and knee radiographs, knee MRI may be useful to identify structural knee pathologies that may (or may not) be relevant for the symptoms.

When should arthroscopic partial meniscectomy (APM) be proposed?

- Surgery should not be proposed as a first line of treatment of degenerative meniscus lesions.
- After 3 months with persistent pain/mechanical symptoms despite appropriate rehabilitation: for a degenerative meniscus with normal X-rays/abnormal MRI (grade III meniscus lesion), APM may be proposed.
- Surgery can be proposed earlier for patients presenting with considerable mechanical symptoms.
- No arthroscopic surgery should be proposed for a degenerative meniscus lesion with advanced osteoarthritis on weight-bearing radiographs.

Most of the published randomised control trials (RCTs) found no difference in terms of clinical outcomes after surgery compared to non-operative treatment.

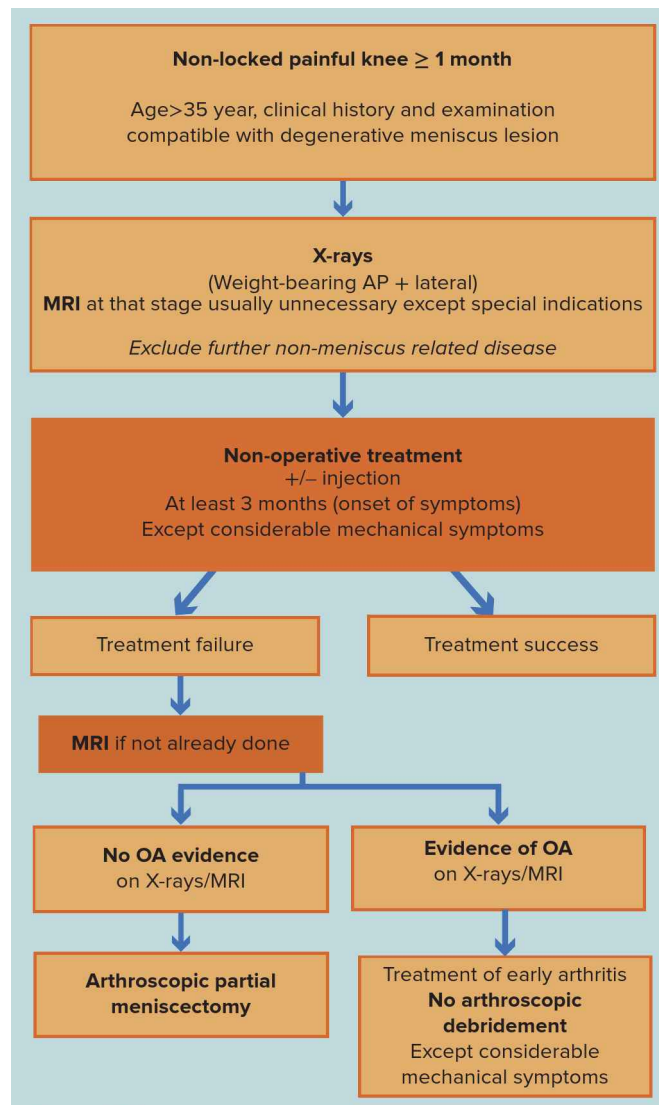


Figure 35.4 ESSKA Meniscus Consensus algorithm

severe symptoms. Pain and restriction of range of motion occur soon after injury. Intermittent locking may occur as a result of the torn flap, the 'bucket handle', impinging between the articular surfaces. This may unlock spontaneously with a clicking sensation. This often occurs in association with ACL tears. In these patients, a history of locking may be due to either the ACL or the meniscal injury.

On examination, the signs of a meniscal tear include:

- joint line tenderness (palpated with the knee flexed at 45–90°)
- joint effusion—this is usually present, although absence of an effusion does not necessarily rule out meniscal damage
- pain—usually present with knee hyperflexion, such as squatting, especially with posterior horn tears
- restricted range of motion of the knee joint—this may be due to the torn meniscal flap or the effusion.

The flexion/rotation (McMurray's) test (Fig. 35.2p) is positive when pain is produced by the test and a clunk is heard or felt that corresponds to the torn flap being impinged in the joint. However, it is not necessary to have a positive McMurray's test (i.e. a clunk) to make a diagnosis of a torn meniscus. The hyperflexion portion of the McMurray's test provokes pain in most meniscal injuries. Pain produced by flexion and external rotation suggests medial meniscal damage, whereas pain on internal rotation indicates lateral meniscal pain. Asking patients where they feel pain during hyperflexion manoeuvres gives a suggestion of the location of the tear, medial or lateral. MRI is the investigation of choice. This can aid management if the MRI shows either a complex tear versus minimal damage or, more rarely, a peripheral meniscus tear.

Treatment

The management of meniscal tears varies depending on the severity of the condition, the age of the patient and the presence of mechanical pathology. At one end of the

spectrum, a small tear or a degenerative meniscus tear, presenting with pain but not with mechanical symptoms such as locking or range of motion (ROM) restriction, should initially be treated conservatively.³⁴

In a highly publicised paper published in the *New England Journal of Medicine* in 2013, the outcomes after arthroscopic partial meniscectomy in patients without knee osteoarthritis but with symptoms of a degenerative medial meniscus tear, were no better than those after a sham surgical procedure.³⁵ A recent trial compared exercise therapy with arthroscopic partial meniscectomy in middle-aged patients with degenerative meniscal tears and found no clinically relevant difference between the two groups in Knee injury and Osteoarthritis Outcome Score (KOOS) at 2 years. At 3 months, muscle strength had improved in the exercise group.³⁶

An algorithm for the management of degenerative meniscal lesions is shown in the box above.

On the other hand, a large painful 'bucket handle' tear, causing a locked knee, requires urgent arthroscopic surgery. The majority of meniscal injuries fall somewhere between these two extremes and the decision on whether to proceed immediately to arthroscopy must be made on the basis of the severity of the symptoms and signs, as well as the demands of the athlete. Table 35.6 provides some clinical guidance for choosing either conservative or surgical treatment.

PRACTICE PEARL

Physical examination features and MRI can predict those patients with early osteoarthritis and a meniscus tear who might benefit from arthroscopic surgery.³⁷ The highest likelihood of benefit from arthroscopic partial meniscectomy came in patients with clinical findings of increasing pain (as distinct from stable pain) as well as locking when these were complemented by MRI findings of a displaced meniscal tear but no marrow lesions.³⁷

Table 35.6 Clinical features of meniscal injuries that may affect prognosis

Factors that may indicate non-surgical treatment is likely to be successful	Factors that may indicate surgery will be required
Symptoms develop over 24–48 hours following the injury	Severe twisting injury, athlete is unable to continue playing
Minimal injury or no recall of specific injury	Locked knee or severely restricted ROM
Able to weight-bear	Positive McMurray's test (palpable clunk)
Minimal swelling	Pain on McMurray's test with minimal knee flexion
Full range of movement with pain only at end of ROM	Presence of associated ACL tear
Pain on McMurray's test only in inner range of flexion	Little improvement of clinical features after 3–6 weeks of non-surgical treatment
Previous history of rapid recovery from similar injury	
Early degenerative changes on plain radiographs	

ROM = range of motion

Non-surgical management of meniscal injuries

Non-surgical management of relatively minor meniscal injuries will often be successful, particularly in the athlete whose sporting activity does not involve twisting activities. The principles of non-surgical management are the same as those following partial meniscectomy (Table 35.7), although the rate of progress may vary depending on the clinical features.

The criteria for return to sport following meniscal injury, treated surgically or non-surgically, are shown below. If appropriate rehabilitation principles have been followed, then the criteria will usually all be satisfied:

- no effusion
- full range of movement
- normal quadriceps and hamstring function
- normal hip external rotator function
- good proprioception
- functional exercises performed without difficulty
- training performed without subsequent knee symptoms
- simulated match situations undertaken without subsequent knee symptoms.

Surgical management of meniscal injuries

The aim of surgery is to preserve as much of the meniscus as possible. Some meniscal lesions are suitable for repair by arthroscopic meniscal suture. The decision as to whether or not to attempt meniscal repair is based on several factors, including acuity of the tear, age of the patient, stability of the knee, and tear location and orientation. The outer one-third of the meniscus rim has a blood supply and tears in this region can heal.

The tear with the best chance of a successful repair is an acute longitudinal tear in the peripheral one-third of the meniscus in a young patient.³⁸ Degenerative, flap, horizontal cleavages and complex meniscal tears are poor candidates for repair.³³ Young patients have a higher success rate of healing the meniscus. Peripheral meniscus tears in otherwise stable knees without concomitant ligament damage have a reduced success rate.³⁹

Partial tears may require removal of the damaged flap of the meniscus. Patients with degenerative tears with no or minimal cartilage wear will be less symptomatic than those patients with concomitant cartilage damage.

Rehabilitation after meniscal surgery

Rehabilitation should always commence prior to surgery and in some cases surgery can be avoided because 'prehabilitation' leads to full recovery. For patients scheduled for surgery it is important to:

- reduce pain and swelling with the use of electrotherapeutic modalities and gentle ROM exercises

- maintain strength of the quadriceps, hamstrings, and hip abductor and extensor muscles
- protect against further damage to the joint (patient may use crutches if necessary)
- explain the surgical procedure and the postoperative rehabilitation program to the patient.

The precise nature of the rehabilitation process will depend on the extent of the injury and the surgery performed (Table 35.7). Arthroscopic partial meniscectomy is usually a straightforward procedure followed by a fairly rapid return to activity. Some athletes with a small isolated medial meniscal tear are ready to return to sport after 4 weeks of rehabilitation.

PRACTICE PEARL

The rehabilitation process usually takes longer if there has been a more complicated tear of the meniscus, especially if the lateral meniscus is injured.

The presence of associated abnormalities, such as articular cartilage damage or ligament (MCL, ACL) tears, will necessarily slow down the rehabilitation process.

The time to return to pre-injury level of competition is significantly longer after lateral than medial meniscectomy in elite professional soccer athletes. Lateral meniscectomy has a higher incidence of adverse events in the early recovery period, including pain/swelling and the need for further arthroscopy. It is also associated with a significantly lower rate of return to play (RTP).⁴⁰

If the athlete returns to play before the knee is properly rehabilitated, he or she may not experience difficulty during the first competition, but may be prone to develop recurrent effusions and persistent pain.

PRACTICE PEARL

A successful return to sport after meniscal knee surgery should not be measured by the time to play the first match. Clinicians must consider the player's longer term performance (season, multiple seasons). Is there acceleration of osteoarthritis?

Close monitoring is essential during post-meniscectomy rehabilitation as the remaining meniscus and underlying articular cartilage slowly increase their tolerance to load-bearing activities. Constant reassessment after progressively more difficult activities should be performed by the therapist monitoring the rehabilitation program. The development of increased pain or swelling should result in the program being slowed or revised accordingly.

Table 35.7 Rehabilitation program for both non-surgical management of meniscal injury and following arthroscopic partial meniscectomy

Phase	Goal of phase	Time post-injury	Physiotherapy	Exercise program	Functional/sport-related activity
Phase 1	Control swelling Maintain knee extension Knee flexion to 100°+ 4/5 quadriceps strength 4+/5 hamstring strength	0–1 week	Cryotherapy Electrotherapy Compression Manual therapy Gait re-education Patient education	Gentle ROM (extension and flexion) Quadriceps/VMO setting Supported (bilateral) calf raises Hip abduction and extension Hamstring pulleys/rubbers Gait re-education drills Light exercise bike	Progress to FWB and normal gait pattern
Phase 2	Eliminate swelling Full ROM 4+/5 quadriceps strength 5/5 hamstring strength	1–2 weeks	Cryotherapy Electrotherapy Compression Manual therapy Gait re-education Exercise modification and supervision	ROM drills Quadriceps/VMO setting Mini-squats and lunges Leg press (double, then single-leg) Step-ups Bridges (double, then single-leg) Hip abduction and extension with rubber tubing Single-leg calf raises Gait re-education drills Balance and proprioceptive drills (single-leg)	Swimming (light kick) Exercise bike Walking
Phase 3	Full ROM Full strength Full squat Dynamic proprioceptive training Return to running and restricted sport-specific drills	2–3 weeks	Manual therapy Exercise/activity modification and supervision	As above—increase difficulty, repetitions and weight where appropriate Jump and land drills Agility drills	Running Swimming Road bike Sport-specific exercises (progressively sequenced) (e.g. running forwards, sideways, backwards, sprinting, jumping, hopping, changing direction, kicking)
Phase 4	Full strength, ROM, and endurance of affected limb Return to sport-specific drills and restricted training and match play	3–5 weeks	As above	High-level sport-specific strengthening as required	Return to sport-specific drills, restricted training and match play

FWB = full weight-bearing; ROM = range of motion; VMO = vastus medialis obliquus

MEDIAL (TIBIAL) COLLATERAL LIGAMENT INJURY

Injury to the MCL usually occurs as a result of a valgus stress to the partially flexed knee. This can occur in a non-contact mechanism such as downhill skiing or in contact sports when an opponent falls across the knee from lateral to medial. MCL tears are classified on the basis of their severity into grade I (mild, first degree), grade II (moderate partial ruptures, second degree) or grade III (complete tears, third degree).

In patients with a grade I MCL sprain, there is local tenderness over the MCL on the medial femoral condyle or medial tibial plateau but usually no swelling. When a valgus stress is applied at 30° of flexion, there is pain but no increased laxity. Ligament integrity is intact.

A grade II MCL sprain is produced by a more severe valgus stress. Examination shows marked tenderness, sometimes with localised swelling. A valgus stress applied at 30° of knee flexion causes pain. Some laxity (typically <5 mm) is present but there is a distinct end point. The knee is stable at full extension; ligament integrity is compromised but intact throughout its length.

A grade III tear of the MCL results from a severe valgus stress that causes a complete tear of the ligament fibres. The patient often complains of a feeling of instability and a 'wobbly knee'. The amount of pain is variable and frequently

not as severe as one would expect given the nature of the injury. On examination, there is tenderness over the ligament and valgus stress applied at 30° of flexion reveals gross laxity without a distinct end point. A minor valgus instability is usually also found at full extension. This test may not provoke as much pain as incomplete tears of the ligament due to complete disruption of the nociceptive fibres of the ligament.

Grade III MCL injuries are frequently associated with a torn ACL. The presentation of medial joint line tenderness and lack of full extension is more a reflection of MCL injury than meniscal pathology. The lateral meniscus is more at risk because the mechanism of injury typically opens the medial side and compresses the lateral side.

While swelling is uncommon in grade 1 sprains, it may occasionally be seen with grade 2 injuries. In grade 3 sprains, there is associated capsular tearing (deep fibres and superficial) and fluid escapes; some degree of swelling is common although a tense effusion is not present.

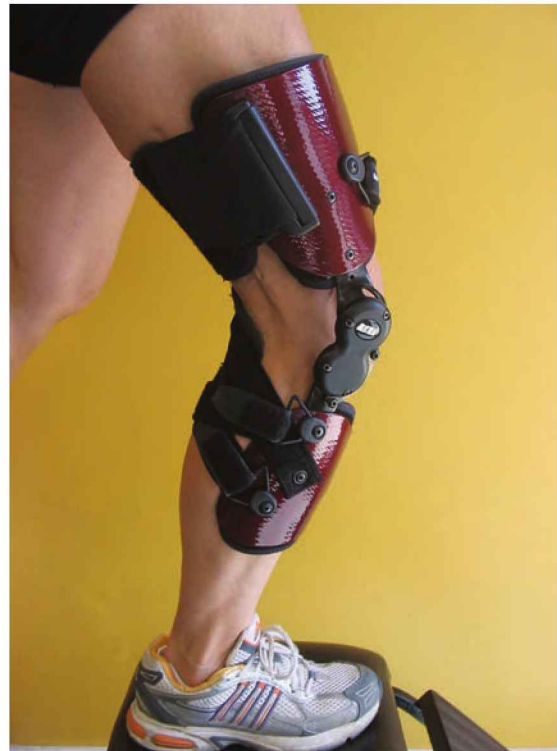
Distal MCL injuries have a tendency to recover more slowly than proximal lesions.⁴¹

Treatment

A hinged knee brace (Fig.35.5a) provides support and protection to the injured MCL for a period of 4–6 weeks, during which time the athlete undertakes a



(a)



(b)

Figure 35.5 Splints (a) Hinged knee brace (b) Limited-motion knee brace

comprehensive rehabilitation program. The brace and the exercise promote early healing of the ligament and any associated capsular injury.

In one study, there were no differences between patients with grade III MCL injuries that were treated non-surgically (i.e. bracing) or surgically, thus non-surgical treatment is recommended.⁴² A typical rehabilitation

program for milder MCL injuries (grade I and mild grade II) is shown in Table 35.8.

The more severe MCL injury (the severe grade II or grade III tear) could be treated with a limited-motion knee brace (Fig.35.5b) and requires a longer period of rehabilitation and a rehabilitation program as shown in Table 35.9.

Table 35.8 Rehabilitation of a mild MCL injury (see Figs 35.5 and 35.6)

Phase	Goal of phase	Time post-injury	Physiotherapy treatment	Exercise program	Functional/sport-related activity
Phase 1	Control swelling Knee flexion to 100°+ Allow +20° extension 4/5 quadriceps strength 4+/5 hamstring strength	0–1 week	Cryotherapy Electrotherapy Compression Manual therapy Gait re-education Patient education	Gentle ROM (flexion mainly) Quadriceps/VMO setting Supported (bilateral) calf raises Hip abduction and extension Hamstring pulleys/rubbers Gait re-education drills	Progress to FWB and normal gait pattern
Phase 2	Eliminate swelling Full flexion ROM Allow +10° extension 4+/5 quadriceps strength 5/5 hamstring strength Return to light jogging	1–2 weeks	Cryotherapy Electrotherapy Compression Manual therapy Gait re-education Exercise modification and supervision	ROM drills Quadriceps/VMO setting Mini-squats and lunges Leg press (double, then single-leg) Step-ups Bridges (double, then single-leg) Hip abduction and extension with rubber tubing Single-leg calf raises Gait re-education drills Balance and proprioceptive drills (single-leg)	With hinged knee brace Straight-line jogging Swimming (light kick) Road bike
Phase 3	Full ROM Full strength Full squat Dynamic proprioceptive training Return to running and restricted sport-specific drills	2–4 weeks	Manual therapy Exercise/activity modification and supervision	As above—increase difficulty, repetitions and weight where appropriate Jump and land drills Agility drills	Progressive running Swimming Road bike Sport-specific exercises (progressively sequenced) (e.g. running forwards, sideways, backwards, sprinting, jumping, hopping, changing direction, kicking)
Phase 4	Full strength, ROM, and endurance of affected limb Return to sport-specific drills and restricted training and match play	3–6 weeks	As above	High-level sport-specific strengthening as required	With hinged knee brace Return to sport-specific drills, restricted training and match play

FWB = full weight-bearing; ROM = range of motion; VMO = vastus medialis obliquus

Table 35.9 Rehabilitation of a moderate-to-severe MCL injury (see Figs 35.6 and 35.7)

Phase	Goal of phase	Time post-injury	Physiotherapy treatment	Exercise program	Functional/sport-related activity
Phase 1	Control swelling Knee flexion to 90°+ Allow +30° extension 4/5 quadriceps strength 4+/5 hamstring strength	0–4 weeks	Limited-motion knee brace (limited 0–30°) Cryotherapy Electrotherapy Compression Manual therapy Gait re-education Patient education	Exercises done in brace Gentle flexion ROM Extension ROM to 30° only Quadriceps/VMO setting Supported (bilateral) calf raises Hip abduction and extension Hamstring pulleys/rubbers Gait drills	Initially NWB/PWB Progress to FWB Walking (normal gait pattern)
Phase 2	FWB Eliminate swelling Full ROM 4+/5 quadriceps strength 5/5 hamstring strength	4–6 weeks	Removal of brace 4–6 weeks Cryotherapy Electrotherapy Compression Manual therapy Gait re-education Exercise modification and supervision	ROM drills Quadriceps/VMO setting Mini-squats and lunges Leg press (double, then single-leg) Step-ups Bridges (double, then single-leg) Hip abduction and extension with rubber tubing Single-leg calf raises Gait re-education drills Balance and proprioceptive drills (single-leg)	Swimming (light kick) Road bike Walking
Phase 3	Full ROM Full strength Full squat Dynamic proprioceptive training Return to light jogging Return to running and restricted sport-specific drills	6–10 weeks	Manual therapy Exercise/activity modification and supervision	As above—increase difficulty, repetitions and weight where appropriate Jump and land drills Agility drills	Straight-line jogging with hinged knee brace (no earlier than 6 weeks) Running Swimming Road bike Sport-specific exercises (progressively sequenced) (e.g. running forwards, sideways, backwards, sprinting, jumping, hopping, changing direction, kicking)
Phase 4	Full strength, ROM and endurance of affected limb Return to sport-specific drills and restricted training and match play	8–10/12 weeks	As above	High level of sport-specific strengthening as required	Return to sport-specific drills, restricted training and match play

FWB = full weight-bearing; NWB = non weight-bearing; PWB = partial weight-bearing; ROM = range of motion; VMO = vastus medialis obliquus



(a)



(c)



(b)



(d)

Figure 35.6 Knee rehabilitation (a) Quadriceps drills— isometric contraction (b) Assisted knee flexion. Place hands behind the thigh and pull the knee into flexion (c) Double-leg calf raise. Progression of the double-leg calf raise should incorporate an increase in range, sets and repetition and speed of movement. The eccentric component should be emphasised (d) Bridging. This is used to develop hamstring, gluteal and core muscular strength



(e)



(f)

Figure 35.6 (cont.) (e) Bridging with Swiss ball. A Swiss ball may be used to progress the exercise (f) Hip extension—with rubber tubing (g) Hip abduction with rubber tubing (h) Rubber tubing eccentric stride catch—standing



(g)



(h)



(i)

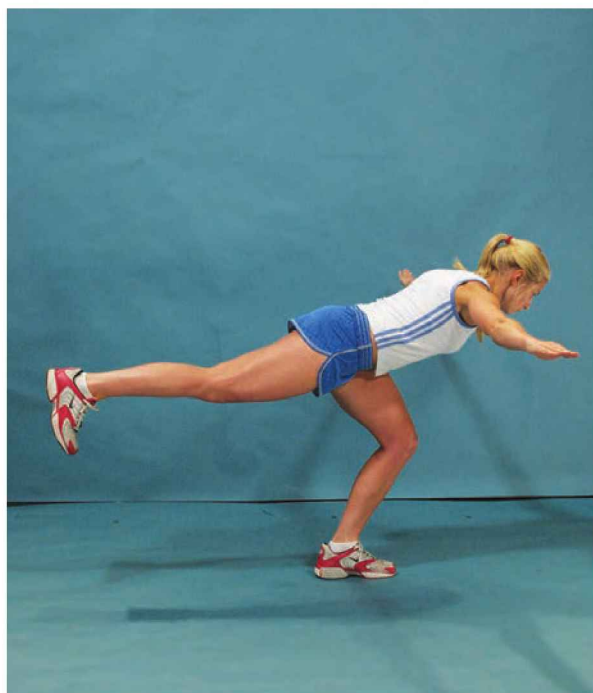


(j)

Figure 35.6 (cont.) (i) Lunge—performed as shown. Progression involves a combination of increasing the number of sets and repetitions, increasing the depth of the lunge and finally by holding additional weight (j) Double-leg quarter squat (k) Single-leg half squat. All squat exercises should be pain-free. The squat may be aided by the use of a Swiss ball. Particular attention must be given to technique and control of the pelvis, hip and knee. Progression of the squat is similar to that of progression of the leg press exercise (l) Arabesque single-leg squat



(k)



(l)



(m)



(n)

Figure 35.6 (cont.) (m) Rebounder—jogging. A good way to introduce higher impact activity. (n) Static proprioceptive hold/throwing ball. Ball throwing or 'eyes-closed' exercises can provide an excellent functional challenge (o) Wobble board (p) Dura disc balance



(o)



(p)



(a)



(b)

Figure 35.7 Functional activities (a) Jump and land from block. This exercise may be used to replicate functional movements in many sports. Begin the exercise from a small height and jump without rotation. This exercise can be progressed by increasing the height of the jump and rotating 90° during the jump (b) Plyometric jumps over block—lateral plyometric exercises should only be included in the later stages of rehabilitation. Each plyometric exercise should be sport-specific



(c)



(d)

Figure 35.7 (cont.) (c) Carioca exercises—crossover stepping exercise (d) Figure-of-eight running

ANTERIOR CRUCIATE LIGAMENT INJURIES

The annual incidence of ACL tears is 81 per 100 000 inhabitants aged 10–64 years in Europe.⁴³ In the United States more than 200 000 ACL reconstructions are performed annually at a cost of over US\$2 billion.^{44, 45}

ACL injuries occur most frequently in sports involving pivoting and sudden deceleration (e.g. football, basketball, netball, soccer, European team handball, gymnastics, downhill skiing). The incidence rate of ACL tears is between 2.4 and 9.7 times higher in female athletes competing in similar activities.^{46–51}

A genetic predisposition to ACL tear has been proposed; nonetheless, although specific gene polymorphisms and haplotypes have been identified, it is difficult to come to a conclusion on the basis of the existing literature. More studies are needed in larger populations of different ethnicities. Gene-gene interactions and gene expression studies in the future may delineate the exact role of these gene polymorphisms in ACL tears.⁵²

Although ACL tears can occur in isolation, this is relatively uncommon. More frequent are those occurring in combination with associated injuries, such as meniscal tears, articular cartilage injuries or MCL injuries.⁵³ Principally, all ACL injuries have associated BMLs that are visible on MRI and form a footprint of the injury mechanism.³⁰

PRACTICE PEARL

As the torn ACL is only one part of a more complex problem in most ACL injured knees, the diagnosis 'ACL injury' may underplay the severity of an acute knee injury.

Anatomy of the ACL

The ACL is one of four main ligaments that stabilise the knee joint and provides nearly 90% of the stability to anterior translation. The femoral insertion site of the ACL is located on the lateral intercondylar wall 43% of the distance from the proximal to distal femoral articular margin and, in the anterior to posterior dimension, 2.5 mm plus the radius of the ACL footprint anterior to the posterior articular margin.⁵⁴ The ACL's insertion on the tibial plateau is 15 mm anterior to the PCL insertion and two-fifths of the width from the medial to lateral tibial spine.⁵⁴

The two bundles that make up the ACL are named according to their tibial insertion. The anteromedial (AM) bundle inserts anteriorly and medially on the tibia, near the medial intercondylar tubercle, while the posterolateral (PL) bundle inserts posteriorly and laterally on the tibia. The fibres of the AM bundle are isometric overall, while the PL bundle fibres are anti-isometric. This means that

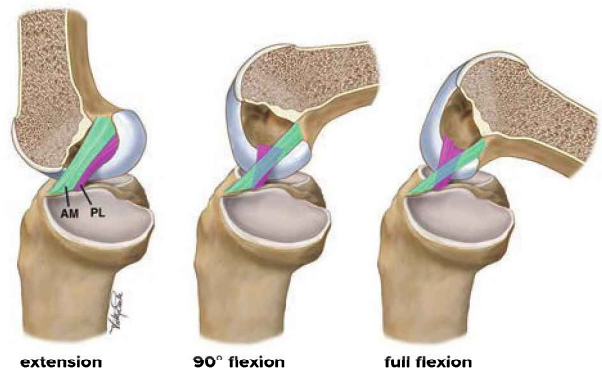


Figure 35.8 ACL bundles during knee flexion. From extension to full flexion, the posterolateral (PL) bundle femoral insertion moves in an arcuate path around the anteromedial (AM) bundle femoral insertion

the length of one portion of the ACL fibres varies during flexion-extension movements.⁵⁵

During flexion, the PL bundle wraps around the AM bundle at the femur; the femoral insertion of the PL bundle forms an arc around the AM bundle's femoral insertion (Fig. 35.8). The anterior fibres of the AM bundle are more isometric, but nevertheless they are less taut between 0° and 30° flexion, which allows them to deform (deformation with superior concavity) as they make contact with the anterior side of the intercondylar notch, thereby allowing full knee extension. Conversely, they are continuously under tension between 30° and 130° flexion.

The posterior fibres of the PL bundle are the least isometric fibres; they are all parallel and completely taut in extension. Between 0° and 90° flexion, they gradually relax and then become taut again beyond 90° flexion. The PL bundle seems to have the most control over tibial rotation due to its more lateral position.⁵⁵

An intact ACL stabilises the femur on the tibia and prevents anterior tibial translation and rotation during agility exercises, jumping, deceleration and pivoting with sudden changes of direction.

Mechanism of ACL injury

In ball sports, two common mechanisms cause ACL tears:

1. a cutting manoeuvre^{56–58}
2. single-leg landing.

Cutting or sidestep manoeuvres are associated with dramatic increases in the varus-valgus and internal rotation moments, as well as deceleration. The typical ACL injury occurs with the knee externally rotated and in 10° to 30° of flexion when the knee is placed in a valgus position as the athlete takes off from the planted foot and internally rotates their upper body with the

aim of suddenly changing direction (Fig. 35.9a).^{59, 60} The ground reaction force falls medial to the knee joint during a cutting manoeuvre and this added force may tax an already tensioned ACL and lead to failure. Similarly, in the landing injuries, the knee is close to full extension.

High-speed activities such as cutting or landing manoeuvres require eccentric muscle action of the quadriceps to resist further flexion. It is hypothesised that vigorous eccentric quadriceps muscle action may play a role in disruption of the ACL. Although this normally may be insufficient to tear the ACL, it may be that the addition of a valgus knee position and/or rotation could trigger an ACL rupture.

One question that is often asked is why the ACL tears in situations and manoeuvres that the athlete has performed many times in the past. Frequently, there is some external factor that renders the athlete susceptible. The athlete could be off balance, be pushed or held by an opponent, be trying to avoid collision with an opponent or have adopted an unusually wide foot position. These perturbations may contribute to the injury by causing the athlete to plant the foot so as to promote unfavourable lower extremity alignment; this may be compounded by inadequate muscle protection and poor neuromuscular control.⁶⁰ Fatigue and loss of concentration may also be relevant factors.

What has become recognised is that unfavourable body movements in landing and pivoting can occur, leading to what has become known as the 'functional valgus' or 'dynamic valgus' knee, a pattern of knee collapse where the knee falls medial to the hip and foot. This has been called by Ireland the 'position of no return' or perhaps it should be termed the 'injury-prone position' since there is no proof that one cannot recover from this position (Fig. 35.9b).⁶¹ Intervention programs aimed to reduce the risk of ACL injury are based on training safer neuromuscular patterns in simple manoeuvres such as cutting and jump-landing activities.

The mechanism of ACL injury in skiing is different from that in jumping, running and cutting sports such as football and basketball. In skiing, most ACL injuries result from internal rotation of the tibia with the knee flexed greater than 90°, a position that occurs when a skier who is falling backwards catches the inside edge of the tail of the ski.⁶² Intervention programs in skiing are aimed at increasing the skier's awareness of patterns that are injurious to the knee and giving alternative strategies in the hope of avoiding these patterns altogether.

Clinical features

The typical features of the history include the following.

- The patient often describes an audible 'pop', 'crack' or feeling of 'something going out and then going back'.

- Most complete tears of the ACL are extremely painful, especially in the first few minutes after injury.
- Athletes are often initially unable to continue their activity. Occasionally pain will limit further activity and this is usually associated with a large tense effusion. This is the clinical feature of a haemarthrosis. Occasionally, swelling is minimal or delayed.
- At times the athlete tries to recommence the sporting activity and feels instability or a lack of confidence in the knee. Occasionally the athlete may resume playing and suffer an acute episode of instability.

Most athletes with an ACL tear present to a sports medicine practitioner between 24 and 48 hours following the injury. At this stage, it may be difficult to examine the knee. The best time to examine a patient with this condition is in the first hour following the injury before the development of a tense haemarthrosis, which limits the examination. After a few days, when the swelling has started to settle and the pain is less intense, the examination becomes easier to perform in most cases.

After ACL rupture, the examination findings are typical:

- athletes have restricted movement
- they may have widespread mild tenderness
- lateral joint tenderness is often present (this is likely due to the impact pain from the collision of tibia and



(a)

Figure 35.9 Abnormal positions that may lead to ACL injury
(a) The typical position during the cutting manoeuvre which leads to ACL injury

REPRODUCED WITH PERMISSION OF BRITISH JOURNAL OF SPORTS MEDICINE

	muscles involved	Position of safety	body position	body position	Point of 'no return'	muscles involved
back			normal lordosis		forward flexed, rotated opposite side	
hips	extensors abductors gluteals		flexed neutral abduction adduction, neutral rotation		adduction internal rotation	flexors adductors iliopsoas
knee	flexors hamstring		flexed		less flexed, valgus	extensors quadriceps
tibial rotation	plantar flexors		neutral		internal or external	dorsiflexors
landing pattern	gastrocnemius posterior tibialis		both feet in control balanced		one foot out of control unbalanced	peroneals tibialis anterior

(b)

Figure 35.9 (cont.) (b) The positions of safety and of 'no return'

femur at the time of injury occurring in the valgus position) resulting in a bone marrow lesion (BML)

- medial joint line tenderness may be present if there is an associated medial meniscus injury.

PRACTICE PEARL

The Lachman's test is positive in ACL disruption and is the most useful test for this condition.

The Lachman's test (Figs 35.2h-k) is a core competency for clinicians who see patients with knee injuries. It is judged on both the degree of anterior translation and the quality of the endpoint.^{3, 63} With an effusion, maximum anterior tibial excursion may be restricted, but the quality of the endpoint is indistinct.

A positive pivot shift (or jerk) test (Fig. 35.2m) is diagnostic of ACL deficiency, but it requires the patient to have an intact MCL and iliotibial band, as well as the ability to extend the knee almost to full extension.³ In cases of acute injuries, especially with associated injury (e.g. meniscal tear), the pivot shift test is difficult to

perform as the patient is unable to relax sufficiently. The anterior drawer test (Fig. 35.2l) is usually positive in cases of ACL tears but is the least specific test.³ It should always be compared with the other side as often there is a degree of anterior tibial translation with this test which is quite variable within the population.

A plain radiograph of the knee should be performed when an ACL tear is suspected. Although radiographs are often normal, it may reveal an avulsion of the ligament from the tibia or a 'Segond' fracture at the lateral margin of the tibial plateau (Fig. 35.10); this has always been regarded as pathognomonic of an ACL rupture.⁶⁴ More recent studies have suggested that the Segond fragment is actually an avulsion of the anterior lateral ligament of the knee (the mid third capsular ligament) and represents a more severe injury than an isolated ACL tear.⁶⁵ Experienced clinicians and radiologists may also detect signs of increased joint fluid on plain radiographs.

MRI may be useful in demonstrating an ACL injury (Fig. 35.11) when the diagnosis is uncertain clinically, and will also demonstrate the typical bone bruise associated with ACL tears (see box below and Fig. 35.11). However, MRI should be used mainly to detect associated meniscal



Figure 35.10 X-ray showing a Second fracture



Figure 35.11 MRI of ACL (circled) showing the precise location of the tear (arrow)

tears and cartilage injuries when these injuries are suspected clinically.

Surgical or conservative treatment of the torn ACL?

The optimal treatment of the torn ACL is not known, and there are several areas of controversy regarding the

What is a bone bruise?

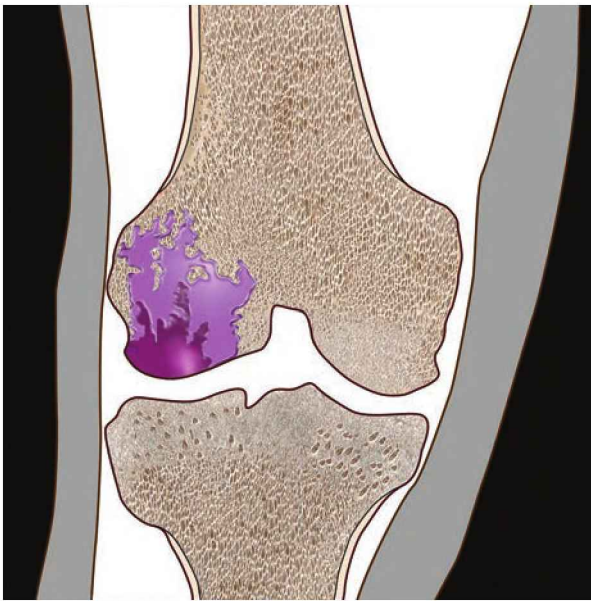
A post-traumatic bone marrow lesion (BML) (commonly referred to as a 'bone bruise') is only visible on MRI and accompanies an ACL injury in 80–98% of cases.^{30, 50} The most common site is the lateral femoral condyle (Fig. 35.12a–c) and the lateral tibial condyle (Fig. 35.12d). The BML is most likely caused by impaction between the posterior aspect of the lateral tibial plateau and the lateral femoral condyle during displacement of the joint at the time of the injury. The presence of a BML indicates impaction trauma to the articular cartilage.³¹

The degree to which BMLs result in permanent injury to the cartilage continues to be investigated. At present it is not clear whether the presence of a BML is significant in the long term. Whether patients with a BML are more prone to osteoarthritis is a hot topic; one report indicated there was no difference in the longer term outcome between those with and without a post-traumatic BML.⁶⁶ Whether the presence of a BML should slow the rehabilitation process is also not clear but most clinicians favour a conservative course of treatment in this regard and limit pounding activities for 3 months post bone bruise.



(a)

Figure 35.12 Bone bruising, evident on MRI, is an important diagnostic and prognostic feature often associated with ACL injury (a) Anteroposterior coronal MRI of a lateral femoral condyle bone marrow oedema in association with an ACL rupture. In radiology exams, this is considered pathognomonic of an ACL rupture until proven otherwise



(b)



(c)

management of ACL injuries. These include: the relative merits of non-surgical versus surgical management; the use of braces to prevent ACL injury or control the ACL-deficient knee; whether surgery should be performed immediately after the injury or should be delayed; whether a delayed reconstruction is to be performed weeks, months or even years after injury; the relative merits of the various surgical techniques; and the benefit of different rehabilitation programs.



(d)

Figure 35.12 (cont.) (b) The pathological appearance of a lateral femoral condyle bone marrow oedema in association with an ACL tear (c) Sagittal MRI appearance of the femoral condyle BML (d) The tibial plateau often also suffers bone marrow oedema; posterolateral tibial bruising (arrowheads in this coronal MRI) is also pathognomonic of ACL injury

There is only one RCT that has compared the results of a surgical and a non-surgical treatment strategy.^{67, 68} In this study, 121 patients aged 18–35 with acute ACL injuries were randomly divided into two treatment groups: rehabilitation plus early ACL reconstruction; or rehabilitation alone with the possibility of a later ACL reconstruction if this was deemed necessary. Professional athletes and those who did not regularly practice sport were excluded from the study whereas competitive athletes on a subprofessional level and recreational athletes were included in the trial. After both 2 and 5 years, there were no differences in terms of knee function reported by the patient, return to sports or surgical treatment of meniscus injury between the groups. However, after 2 years 39% and after 5 years 51% of the group treated with initial rehabilitation needed to have an ACL reconstruction.

This study is used by the proponents of non-surgical rehabilitation treatment with an option of later reconstruction stating that the results at 2 years and 5 years were the same as the group who underwent early surgical reconstruction.^{67, 68} The proponents of early surgical reconstruction use the 51% of the non-surgical group who required later reconstruction by 5 years as support for their claim as well as arguing that the delay in

reconstruction may lead to an increased risk of associated injuries of the meniscus and/or articular cartilage.⁶⁹

Well-structured and extensive rehabilitation seems to be crucial for patients with ACL injury. Still, rehabilitation only is not sufficient for some 50% who are likely to need surgery in order to cope with their injury. Little is known about individual factors related to a good treatment outcome and thus the decision on selecting patients for surgery or rehabilitation is difficult.

Early phase instability at rest (i.e. within the first 3 months after injury) was not a good predictor for the need for later surgical treatment, whereas instability after 3 months was a relatively good prospective predictor.⁷⁰ The following factors should be considered when deciding whether or not to take the surgical option:

- the age of the patient
- the degree of instability in function
- a repairable meniscus tear
- associated knee injuries (e.g. MCL tear, meniscal tear)
- the patient's desire to return to jumping and pivoting sports
- the patient's occupation (e.g. firefighter, police).

The use of functional tests to help 'predict' those that may need operative ACL treatment is currently being investigated. Early reports suggest performing screening examination after 10 sessions of progressive exercise therapy was of value in helping to inform decision making for ACL reconstruction. Those who had surgery were also younger and had a higher activity level.⁷¹

A systematic review of the factors related to the need for surgical reconstruction after ACL rupture reviewed seven studies, three of which were of high quality.⁷² Based on these studies, neither sex (strong evidence) nor the severity of knee joint laxity (moderate evidence) can predict soon after ACL injury whether a patient will need ACL reconstruction following non-operative treatment. All other factors identified in this review either had conflicting or only minimal evidence as to their level of association with the need for surgical reconstruction.

Still, it is important to note that there is limited evidence in determining who will benefit from what treatment. Most decisions are made in consensus between the treating clinician and the patient and are based on empiric knowledge rather than scientific evidence.

The degree of instability may be assessed by a number of parameters. It is important to note that instability at rest (i.e. at clinical examination of the knee) is not a reliable indication of the functional instability experienced by the patient. However, as recurrent episodes of giving way indicate functional instability, this is likely to be a reliable indication of disability and an increased likelihood of needing surgery.

PRACTICE PEARL

For patients presenting with a history of 'giving way' or a history of instability in function despite adequate strength, an ACL reconstruction is recommended.

Many surgeons would advocate ACL reconstruction in patients with concomitant meniscal injuries (50% prevalence) that could be repaired. The decision is also influenced by the demands placed on the knee. A young athlete who wishes to return to a pivoting sport, such as football or basketball, is more likely to need an ACL reconstruction than an athlete who is prepared to confine activity to those sports that do not involve a large amount of twisting, turning and pivoting. However, some of the patients in this category do perform well after 2 years, also after rehabilitation alone.⁶⁷ It is likely that repeated episodes of giving way increase the risk of developing knee osteoarthritis but there is no scientific evidence to support that an ACL reconstruction reduces the incidence of future osteoarthritis. A recent report did not reveal any differences in the frequency of osteoarthritis between surgically reconstructed and non-surgically treated patients 10 years after ACL injury.⁷³

Another important factor to assess is the likelihood of the patient adhering to a comprehensive, time-consuming rehabilitation program. If the patient indicates a lack of willingness to undertake appropriate rehabilitation, treatment may not be successful. It is important to also note that the results of an ACL reconstruction are dependent on a successful postoperative rehabilitation. Other factors to consider are the cost of surgery, rehabilitation and the amount of time off work.

In summary, surgery is generally recommended for athletes wishing to participate in a high-speed sport with constant change of direction and pivoting. Rehabilitation without surgery, however, has been shown to be an alternative for this patient category.⁶⁷ A trial of non-surgical management does not rule out the possibility of eventual later surgery when indicated.

Surgical treatment

There are numerous surgical techniques used to treat ACL injuries. As ACL tears are usually in-substance tears and therefore not suitable for primary repair, reconstruction of the ACL (ACLR) is the surgical treatment of choice. ACLRs are performed 'arthroscopically aided' through small arthroscopic incisions (see box below). Arthroscopic surgery utilises small incisions to help visualise the inside of the joint and facilitates tunnel placement of the ACL graft. Depending on the type of graft, incisions to harvest the graft and secure tunnel access for graft fixation will be made as well.

What happens during ACL reconstruction surgery?

The surgical reconstruction technique involves harvesting the tendon (patellar or hamstring, Fig. 35.13a) through a small incision and threading the tendon through tunnels drilled in the bones. The most crucial parts of the operation are locating the points of entry of the tibial and femoral tunnels and then the fixation of the graft.

The tibial attachment should be in the centre of the previous anterior cruciate attachment (at the level of the inner margin of the anterior portion of the lateral meniscus). The femoral attachment is to the position of optimal or maximised

isometry. This is a position in the intercondylar notch on the femur at which the graft is at a fixed tension throughout the range of knee movement.

Once the graft attachment areas have been delineated and prepared, the graft is fixed by one of a variety of different methods. These methods include interference screw fixation (Fig. 35.13b), staples or the tying of sutures around fixation posts. The better the quality of graft fixation, the more comfortable one is in advancing rehabilitation in the first weeks after surgery.

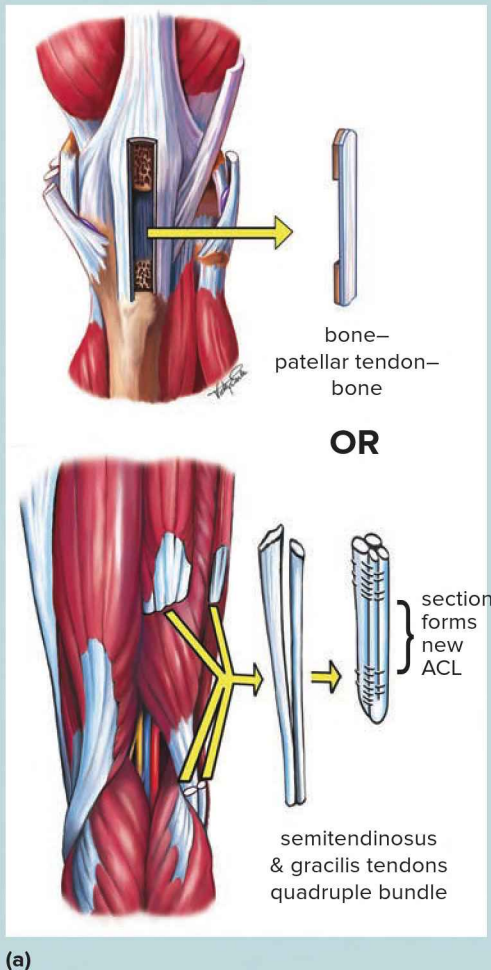
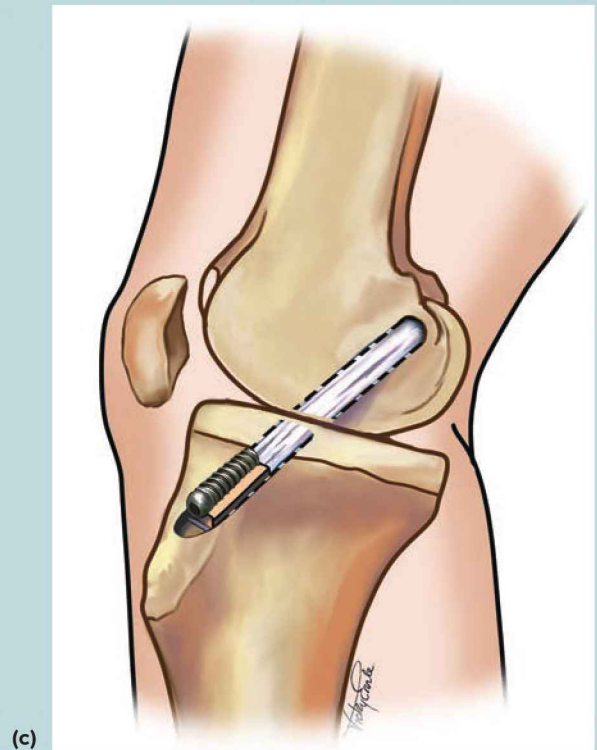


Figure 35.13 The key steps in the process of ACL reconstruction (a) Harvesting graft tissue for the patellar tendon (top panel) or semitendinosus and gracilis tendon ('hamstring graft') ACL reconstruction (b) Replacing the ruptured ACL with the graft tendon tissue; interference screw shown (c) After surgery—the knee with the new graft or 'neoligament' in place



Type of graft

The aim of an ACLR is to replace the torn ACL with a graft that reproduces the normal kinetic functions of the ligament. In most cases, an autogenous graft, taken from around the knee joint, is used. The most common grafts used are the bone-patellar tendon-bone (BTB) autograft involving the central third of the patellar tendon or a multi-stranded hamstring (semitendinosus +/- gracilis tendons) graft using the ipsilateral limb. Other autograft choices include the quadriceps tendon and autografts from the contralateral limb. Allografts (donated tissue from cadaveric specimens) have also been used with the potential benefit of reducing harvest site morbidity but introducing questions of graft incorporation or disease transmission (e.g. HIV, hepatitis, bacterial). Graft choice is dependent on a number of factors including surgeon competency and familiarity with the various techniques.

Patellar tendon versus hamstring tendon graft

There is debate on graft choice, in particular patellar tendon versus hamstring tendon. Reviews of the numerous research papers comparing the two techniques have shown very little difference in outcome measures and there is no universal agreement on which technique is superior.^{74, 75} Each case should be considered on its merit, taking into account some of the differences in potential postoperative issues.

For example, after patellar tendon reconstruction, pain with kneeling is common and up to 50% of patients report anterior knee pain with kneeling. Patients who have a hamstring graft ACL reconstruction have decreased end-range knee flexion power. The incisions used for hamstring grafts are smaller and more cosmetic. Potential problems need to be addressed in the rehabilitation program and for that reason we advocate rehabilitation regimens with different emphasis for the two types of surgery.

Autograft versus allograft

There has been considerable debate regarding the relative merits of autograft (using the patient's own tissue) and allograft (using another person's tissue). A summary of the nine RCTs and 10 systematic reviews showed statistically significant differences in favour of autograft for clinical failure, the Lachman test, the instrumented laxity test and the Tegner score.⁷⁶ When subgroup analyses were conducted based on whether irradiation was used, autograft achieved better clinical outcomes than irradiated allograft in terms of the Lysholm score, clinical failure, the pivot shift test, the Lachman test, the instrumented laxity test and the Tegner score. In addition, there were no significant differences between the autograft and non-irradiated allograft groups for all eight indices. The authors concluded that autograft had greater advantages than irradiated allograft with respect to function and stability,

whereas there were no significant differences between autograft and non-irradiated allograft.

Single bundle versus double bundle

The ACL has two distinct bundles: the anteromedial (AM) bundle and the posterolateral (PL) bundle. It is possible to reconstruct the bundles individually (anatomic double-bundle ACL reconstruction) particularly in large knees. Double-bundle ACL reconstruction (DBACLR) restores the native footprint of the ligament and is definitely superior to isometric ACLR. Whether or not double-bundle anatomic is better than single-bundle anatomic is as yet undecided.

Meta-analyses have suggested that double bundle may result in superiority in rotational stability, the degree of osteoarthritis (OA) changes, and subjective function score postoperatively compared with the single-bundle procedure⁷⁷ and less anterior laxity.⁷⁸ However, there was no difference on functional outcomes between the treatment procedures at long-term follow-up. Given the lack of conclusive evidence of superiority and the doubling of implant costs involved with double-bundle ACLR, single bundle is probably the procedure of choice at this point in time.

Timing of surgery—early or late

The timing of ACL reconstruction after an acute injury has come under review. Traditionally and with very little evidence in support, ACLRs were performed as soon as practical after the injury. However, there is some evidence that delaying the surgery may decrease the postoperative risk of arthrofibrosis (see 'Soft tissue stiffness').⁷⁹ Initial reports suggested 3 weeks as the appropriate delay in surgery but now focus is more on the condition of the knee rather than the actual time. The injured knee should have little or no swelling and have near full ROM; the patient should have a normal gait.

A systematic review examined 22 articles (3583 patients).⁸⁰ Eight articles promoted early reconstruction, whereas the majority of articles found no difference in outcome between early and delayed surgery. Two articles were inconclusive. The authors concluded that there were few or no differences in subjective and objective outcomes related to timing of ACL reconstruction.

Combined injuries

Injuries of the ACL rarely occur in isolation and thus most of all ACL injuries occur in combination with other injuries. The presence and extent of associated injuries may affect the way in which the ACL injury is managed. Associated injury to the MCL (grades II-III) poses a particular problem due to the tendency to develop stiffness after this injury. Most orthopaedic surgeons would initially treat the MCL injury in a knee brace for a period of 4-6 weeks, during

which time the athlete would undertake a comprehensive rehabilitation program (see below). This allows for early healing of the ligament injury and any potential capsular injury, reducing the risk of chronic valgus instability.

ACL injuries are frequently associated with chondral damage. The majority of the literature supports the clinical relevance of cartilage lesions which are correlated with a poorer outcome after ACL reconstruction.⁸¹

Injuries to the posterolateral corner which may occur in conjunction with an ACL injury are considered later in the chapter.

Rehabilitation after ACL injury

Traditionally, most rehabilitation principles have been evaluated as postoperative rehabilitation protocols due to the frequent use of ACL reconstruction as the treatment of choice. However, rehabilitation alone, performed according to a protocol similar to that used postoperatively, was recently shown to provide similar 2-year outcomes as the combination of ACL reconstruction plus rehabilitation.⁶⁷ Thus, rehabilitation may be performed in a similar setting regardless of whether or not additional reconstructive surgery is performed.

Rehabilitation after ACL injury has changed dramatically over time, resulting in greatly accelerated rehabilitation programs.⁸² The speed at which an ACL injury is rehabilitated must take into account the time it takes for soft tissues to heal and remodel as well as the neuromotor control and coordination of the athlete. The most important aims are: reduction of pain, swelling and inflammation; and regaining ROM, strength and neuromuscular control.⁸³ The major change over the past few years is the incorporation of a core stability program along with increased emphasis on proprioceptive and balance exercises. These exercises have been used in successful ACL prevention programs (see Chapter 12). Despite the widespread acceptance of these elements into rehabilitation programs, the two RCTs have not shown convincing evidence of their efficacy.^{84, 85}

The rehabilitation program is shown in Table 35.10. The time frames in the table are a guideline only and must be adjusted depending on the progress of the individual patient. It is essential to rehabilitate each patient individually, taking into consideration the extent of damage to the knee (e.g. articular cartilage damage), the patient's adherence to the exercise program, the amount of knee stiffness, which varies considerably between patients, and the eventual functional aims of the patient (e.g. daily activities, high-level sport). The patient must be taught to monitor the signs and symptoms around the knee following each work-out. Ice may need to be applied if pain, inflammation or swelling is present.

The timing of return to sport (see next section) is dependent on several different factors, including knee

stability, neuromotor function, the nature of the sport, the therapist's and coach's opinion, and the confidence of the patient.

Neuromotor function and muscle strength should be regularly assessed during rehabilitation and prior to return to play. An excellent functional test that can be easily conducted in the clinical setting during the later stages of rehabilitation is the single-leg hop test. The patient performs the single-leg hop by hopping as far as possible using the injured leg, with distance being measured. Athletes with good function are able to land confidently, and 'stick it'. Those with functional disability step further or take another small hop. Comparison should be made with the uninjured leg. Another way of testing function is by incorporating sport-specific drills. Isokinetic testing may be used to evaluate muscle strength. Quadriceps and hamstring strength should approximate that of the uninjured leg. In the light of all these factors and the varying progress of different athletes and the sport to which they are returning, the time for return to sport after ACL injury may vary from 4–12 months.

The use of a brace on return to sport is not necessary but may help the athlete's confidence. Several systematic reviews and RCTs on the topic do not recommend the use of postoperative brace after ACLR. Bracing does not seem to help with pain, function, rehabilitation and stability.⁷⁴

The research into the effectiveness of various rehabilitation techniques has generally been of poor quality and thus limited conclusions can be drawn. The Orthopaedic Section of the American Physical Therapy Association publishes clinical practice guidelines, including the level of supporting evidence, for treatment of knee sprains based on an international classification of functional disability.⁸⁶ The MOON Group (a collection of orthopaedic surgeons) has developed, in conjunction with their physiotherapists, rehabilitation guidelines based on a systematic review of the available evidence.⁸⁷

Considerations in rehabilitation after ACLR

When performed after ACLR, management principles have changed as surgical techniques have changed. There is a better understanding of the initial graft strength and the strength of various graft fixation techniques. There is no difference in joint laxity or clinical outcome between those who underwent accelerated rehabilitation compared to those with a non-accelerated program at 2 years post-surgery.⁸⁸

Without an open arthrotomy, the extensor mechanism has been better preserved with reduced joint adhesions. The principle of complete immobilisation has been replaced with protected mobilisation, with a resultant dramatic decrease in stiffness and increase in ROM of the knee joint. This has allowed earlier commencement of a strengthening program and rapid progression to functional

Table 35.10 Rehabilitation following ACL reconstruction (see Figs 35.5 and 35.6)

Phase	Goal of phase	Time post-surgery	Physiotherapy treatment	Exercise program	Functional/sport-related activity
Prehabilitation (preoperative rehabilitation)	No/minimal swelling Restore full ROM, particularly extension General 4+/5 lower limb strength or better Patient education—anatomy, surgical procedure, rehabilitation commitment, and goal setting	NA	Cryotherapy Electrotherapy Compression Manual therapy Gait re-education Exercise modification and supervision	Dependent on ability of patient In early stages, follow the exercise program from phase 1 and progress to phase 2 If patient has high level of function, start with exercise program from phase 2 and progress weights and repetitions as appropriate	Walking Bike riding Swimming (light kick and no breaststroke)
Phase 1	PWB–FWB Eliminate swelling 0–100° ROM 4+/5 quadriceps strength 5/5 hamstring strength	0–2 weeks	Cryotherapy Electrotherapy Compression Manual therapy Gait re-education Patient education	Gentle flexion ROM Extension ROM to 0° Quadriceps/VMO setting Supported (bilateral) calf raises Hip abduction and extension Hamstring pulleys/rubbers Gait drills	Nil
Phase 2	No swelling Full knee hyperextension Knee flexion to 130°+ Full squat Good balance and control Unrestricted walking	2–12 weeks	Cryotherapy Electrotherapy Compression Manual therapy Gait re-education Exercise modification	ROM drills Quadriceps/VMO setting Mini squats and lunges Leg press (double-leg, then single-leg) Step-ups Bridges (double-leg, then single-leg) Hip abduction and extension with rubber tubing Single-leg calf raises Gait re-education drills Balance and proprioceptive drills (single-leg)	Walking Exercise bike

Table 35.10 (cont.)

Phase	Goal of phase	Time post-surgery	Physiotherapy treatment	Exercise program	Functional/sport-related activity
Phase 3	Full ROM Full strength and power Return to jogging, running, and agility Return to restricted sport-specific drills	3–6 months	Manual therapy Exercise/activity modification and supervision	As above—increase difficulty, repetitions, and weight, where appropriate Jump and land drills Agility drills	Straight-line jogging Swimming (light kick) Road bike Straight-line running at 3 months Progressing to sport-specific running and agility (progressively sequenced) (e.g. running forwards, sideways, backwards, sprinting, jumping, hopping, changing directions, kicking)
Phase 4	Return to sport	6–12 months	As above	High-level sport-specific strengthening as required	Progressive return to sport (e.g. restricted training, unrestricted training, match play, competitive match play)

FWB = full weight-bearing; PWB = partial weight-bearing; ROM = range of motion; VMO = vastus medialis obliquus

exercises. The average time for rehabilitation after ACLR to return to sport has been reduced from around 12 months to 6–9 months before the injured individual could regain sporting activity. The importance of early return to sports in relation to later osteoarthritis development needs to be determined.

Rehabilitation must commence from the time of injury, not from the time of surgery, which may be days, weeks or months later if needed at all.⁸⁹ The initial management aims to reduce pain, swelling and inflammation, thus reducing the amount of intra-articular fibrosis and resultant loss of ROM, strength and function. Treatment should commence immediately after injury, including interferential stimulation, ultrasound and transcutaneous electrical nerve stimulation (TENS), as well as strengthening exercises for the quadriceps, hamstring, hip extensor, hip abductor and calf muscles. Pain-free ROM exercises should also be performed.

Additional muscle strengthening and neuromuscular rehabilitation prior to surgery can improve outcomes.^{90, 91} This is often referred to as 'pre-habilitation' or 'pre-hab'. This period is also an opportunity for the clinician to explain the hospital protocol and the progression and goals of the rehabilitation program. The therapist should set a realistic goal, taking into consideration the individual patient. It is helpful to provide a written explanation as well. If necessary, the postoperative knee brace should be fitted and the use of crutches taught.

Immediately following surgery, weight-bearing status is largely determined by concomitant injuries (e.g. meniscal repair, intra-articular fracture or function of other ligaments). Isolated ACLRs are typically treated as weight-bearing as tolerated, using a brace and/or crutches until adequate quadriceps muscle strength is restored. Instructions should be given regarding the use of crutches as the patient will progress from limited weight-bearing to full weight-bearing for the first 2 weeks.

The rehabilitation programs for patellar tendon and hamstring tendon graft ACLRs are slightly different due to the need to prevent the particular complications associated with each type of reconstruction. Potential problems with the patellar tendon graft are kneeling pain, patellar tendinopathy and/or reduced patellar mobility (see the next section). Therefore, attention must be paid to this area during the rehabilitation program with the use of soft tissue therapy to the patellar tendon, accompanied by a strengthening program for the tendon and patellar taping (Chapter 35) to prevent patellofemoral and fat pad problems. The hamstring graft should be treated as though the patient has had a hamstring tear (Chapter 34), with an appropriately paced rehabilitation program to restore full ROM and strength.

Problems encountered during ACL rehabilitation

Problems that are not treatment dependent

Low back pain

Low back pain is not uncommon in the early stages of the rehabilitation program possibly due to the use of crutches, altered gait patterns or altered sleep patterns. It usually occurs in patients who have a prior history of low back pain.

Problems in non-surgically treated patients

Instability

The main problem that may occur in the rehabilitation of non-surgically treated patients is remaining functional instability. All non-surgically treated patients should have a scheduled appointment with the treating clinician within the first 3 months after injury. When having complaints on symptomatic instability or lack of trust in their knee due to instability despite a successful rehabilitation, an ACLR could be recommended. This is especially true for patients still wishing to resume sporting activities.

Symptomatic meniscus tears

Another problem that may increasingly occur among this group is a symptomatic meniscus tear. Unlike those undergoing ACLR, the non-surgically treated patient does not routinely undergo surgery for meniscus tears associated with their initial injury. Some of these meniscus tears remain asymptomatic, some might heal but some will develop to be symptomatic. Here it is important to differentiate between symptoms adhering to the injured meniscus and symptoms related to the ACL injury and to treat that symptomatic injury. When such problems are encountered, the best option is to let an experienced orthopaedic surgeon assess the injury and discuss treatment alternatives with the patient.

Problems in surgically treated patients

Apart from surgical complications (e.g. infection, deep venous thrombosis, graft harvest site morbidity), a number of secondary problems may occur during the postoperative rehabilitation process.

Patellar region pain

Patellofemoral pain may occur on the injured or the uninjured side. Patients may present with typical symptoms of patellofemoral pain (Chapter 36) but often will not comment on the presence of anterior knee pain as they assume that it is part of the normal process following surgery. The patient should always be asked about symptoms at the front of the knee and the patellofemoral joint should be examined at each visit.

Such factors may be the same as those that predispose a person without ACLR to having patellofemoral pain (Chapter 36), such as altered patellofemoral alignment, altered lower limb mechanics (including internal hip rotation and adduction, apparent knee valgus), quadriceps and hip muscle insufficiencies; or may have developed as part of the ACL injury and reconstruction process. For example, following ACLR, some people have fat pad thickening and oedema, often associated with patella baja or alta, insufficient restoration of quadriceps and hip muscle strength and control, altered tibial rotation (relative to the femur), or persistent loss of knee extension or flexion range. Some of these factors may be aggravated by persistent swelling and/or pain. Such factors should be evaluated and treated as appropriate.

Closed chain activities such as squats and lunges are preferred to open chain activities, as open chain loaded knee extension results in larger patellofemoral forces in the functional knee flexion range of 0° to 60°. ⁹² To minimise patellofemoral force, open chain exercises should be performed between 60° and 90° of knee flexion, while closed chain exercises should be maintained to <60° flexion. However, loaded knee extension at knee flexion angles <60° is thought to strain the ACL graft; clinicians need to exercise caution during early strengthening interventions in this knee range. ⁹³ Notably, patellofemoral problems occur not only with patellar tendon graft reconstructions but also with hamstring tendon graft reconstructions with considerable morbidity. ⁹⁴

The infrapatellar fat pad may be damaged by the arthroscope and can be the source of considerable discomfort after ACL reconstruction. Taping techniques (Chapter 36) can be used to unload the fat pad.

Another complication seen after patellar tendon ACL reconstruction is inferior displacement of the patella (patella baja) due to traction on the patella by tight infrapatellar soft tissue structures. Patellar tendinopathy (Chapter 36) is also seen following ACL reconstruction, especially with patellar tendon grafts.

A common and as yet unexplained finding in cases of chronic ACL insufficiency and following reconstruction is trochlea chondral damage. Factors related to the development and persistence of chondral damage in the tibiofemoral and patellofemoral compartments are currently being studied. ^{95, 96}

Lower limb stiffness

Stiffness in the foot and ankle commonly occurs as a result of a period of non weight-bearing and the wearing of a brace. Tightness of the Achilles tendon is common. These problems may not be recognised until the patient returns to running. Full ROM of these joints should be maintained early in the rehabilitation program with mobilisation and stretching in addition to active plantarflexion/dorsiflexion exercises.

Soft tissue stiffness (arthrofibrosis following surgery)

The rehabilitation program and its rate of progression will be influenced by the intrinsic tissue stiffness or laxity of the patient. This depends on the nature of the patient's collagen and appears to correlate with generalised ligamentous stiffness or laxity throughout the body. Patients with stiff soft tissues may develop a large bulky scar with adhesions after ACLR. These patients are usually slow to regain full flexion and extension and the knee may require passive mobilisation by the therapist. Patients tend to have tight lateral structures around a stiff patellofemoral joint. This is known as arthrofibrosis or stiff knee syndrome. ⁹⁷

Treatment involves encouraging active movement, early passive mobilisation, soft tissue therapy and encouraging early activity. Efforts to control swelling are critical. It may be helpful to remove the brace earlier than usual in these patients. Severe cases may require arthroscopic scar resection as well as a vigorous rehabilitation program.

As mentioned previously, delaying the surgery until all signs of the haemarthrosis have resolved and full ROM has been regained in particular full extension, reduces the incidence of arthrofibrosis in ACL reconstructions.

Soft tissue laxity

The group of patients classified as having 'loose' soft tissue are characterised by generalised increased ligamentous laxity. These patients tend to rapidly gain good ROM in extension and flexion. They are treated by prolonging the time in the brace to prevent hyperextension and restricting the range available. ROM exercises are discouraged, mobilisation contraindicated and full extension work reduced to avoid stretching the graft. The rehabilitation program is slowed in these patients to allow time for the graft to develop as much scar tissue as possible.

Muscle weakness

Muscle weakness occurs following ACL reconstruction despite significant regeneration of the tendon used for the graft (hamstring or patellar). ^{98, 99} A systematic review found that the majority of studies could demonstrate specific muscular imbalances in the ACL-reconstructed leg with control or the contralateral leg with the knee flexors and extensors involved. ¹⁰⁰ In addition, some recent studies have also shown deficits of the hip muscles after ACLR.

Muscular deficits are pronounced within the first 6 months after surgery, but they can persist up to 2 years and longer. The muscular deficits after ACLR are graft related. Extensor deficits are associated with patellar tendon grafts; flexion deficits are associated with hamstring grafts. There is limited evidence with inconsistent study results that there is a difference in postoperative flexor strength between patients with semitendinosus tendon autograft and semitendinosus/gracilis tendon autograft.

The authors of the systematic review recommended that isokinetic examination of flexor and extensor strength should be used as one criterion to decide if an athlete can be allowed to return to unrestricted sporting activities and that the rehabilitation protocol after ACLR should focus on the used graft and aim on negotiation of graft-related muscle weakness. Training to improve hip muscle strength may help to prevent a dynamic valgus during landing positions.¹⁰⁰

Proprioception

Postural control is impaired following ACLR. A systematic review included 10 studies evaluating 644 participants at a mean 29 months follow-up.¹⁰¹ In static balance tasks, there was a trend towards improved postural control in the control group for eyes-open but not eyes-closed conditions. Only four studies evaluated dynamic balance and the results from these were somewhat mixed. They concluded that there appeared to be a trend towards impaired static and dynamic postural control in patients following ACLR surgery, although the limited number of studies and differing methodologies made conclusions tentative.

Patients with ACL injury have poorer proprioception than people without such injuries when measured using joint position sense and threshold to detect passive motion techniques respectively. Patients had poorer proprioception in the injured than uninjured leg and the proprioception of people whose ACL was repaired was better than those whose ligament was left unrepaired.¹⁰²

A review of 22 studies covering a total of 479 patients with a mean age of 27.3 showed that patients after ACLR have altered gait patterns that can persist for up to 5 years after surgery.¹⁰³

Psychological problems

Psychological responses are important factors in RTP after ACLR.¹⁰⁴ In one study, 56 out of 187 athletes (31%) had returned to their pre-injury level of sports participation at 12 months. Significant independent contributions to returning to the pre-injury level by 12 months after surgery were made by psychological readiness to return to sport, fear of re-injury, sport locus of control and the athlete's estimate of the number of months it would take to return to sport, as measured preoperatively and at 4 months postoperatively.

Another study looked at fear of injury and found that athletes participating in sport 2-7 years following their ACL reconstruction generally appear to do so without fear of re-injury.¹⁰⁵ However, they commented that gender, the timing of surgery following injury and the level of sport the athletes returned to may be associated with fear of re-injury following surgery.

Patient psychological factors are predictive of ACLR outcomes.¹⁰⁶ Self-confidence, optimism and self-motivation are predictive of outcomes, which is consistent with the theory of self-efficacy. Stress, social support and athletic self-identity are predictive of outcomes, which is consistent with the global relationship between stress, health and the buffering hypothesis of social support.

In a review of psychosocial factors influencing the recovery of athletes with ACL injury a high internal health locus of control and a high self-efficacy were useful cognitive factors to facilitate the recovery.¹⁰⁷ Athletes with a low level of fear of re-injury had the best knee outcome after the injury followed by a reconstruction. In addition, athletes who returned to sport had less fear of re-injury, and were more experienced and established athletes compared with athletes who did not return to sport. Furthermore, researchers showed that there was a positive relation between goal setting and adherence, which in turn yielded a positive relation with the outcome of the rehabilitation of an ACL injury. There were several psychosocial interventions that appeared to be facilitating the rehabilitation process.

Outcomes after ACL treatment

While the general consensus among the surgical and sporting communities is that those sustaining an ACL injury can and often do make a full recovery after ACLR surgery, research findings suggest otherwise. Four main outcome measures are used to determine the success or failure of ACL treatment:

1. the patient's own perspective of (self-reported) knee function
2. return to pre-injury sporting activities
3. re-injury rate
4. prevalence of osteoarthritis.

Self-reported knee function

The recommended assessment of outcome after ACL injury has changed over the last 15 years. Traditionally, outcome was obtained by an observer, often using a scoring scale such as the Lysholm Knee Scoring Scale,²² the Cincinnati Knee Ligament Rating Scale¹⁰⁸ or the International Knee Documentation Committee (IKDC).¹⁰⁹ It is well known that treating surgeons underestimate symptoms and overestimate function when compared to the patient's own opinion. This was also confirmed after ACL reconstruction in a study from 2001.¹¹⁰ Thus, the patients' own perspective of knee function has been in focus over the last two decades and the use of validated self-administered questionnaires has been promoted. Patient-reported outcomes are generally good at a minimum of 10 years following ACL reconstruction.¹¹¹

After ACLR, data from large samples could be obtained from registries, mainly from Sweden, Norway and Denmark. Here, a large improvement could be seen over the first 2 years after surgery, suggesting that the self-reported knee function is greatly improved by treatment. However, a report from the Swedish ACL registry with data from more than 2000 individuals reported that self-reported knee-related quality of life, knee pain and knee symptoms were much worse after ACL reconstruction¹¹² as compared to an age-matched community-based sample.¹¹³

Little is known about self-reported knee function after non-surgical treatment. In an RCT comparing surgical and non-surgical treatment strategies, no differences were found between the two treatment arms after 2 and 5 years.^{67, 68} The results were comparable to those previously reported after ACLR.

Eleven studies reported quality of life (QoL) in 473 ACL-deficient individuals, at a mean of 10 (range 5–23) years following ACL rupture.¹¹⁴ Eight studies reported knee-related QoL using the Knee injury and Osteoarthritis Outcome Score QoL subscale (KOOS-QoL); scores were impaired compared to population norms. Health-related QoL, measured with the SF-36 domain scores in five studies, was similar to population norms, but impaired compared to physically active populations. Meta-analysis revealed no significant differences in KOOS-QoL and SF-36 scores between ACL-deficient and ACL-reconstructed groups.

In summary, both surgical and non-surgical treatment has been shown to improve self-reported knee function after ACL injury. There exists no scientific evidence to support superiority of one treatment over the other. Two years after injury, ACL-injured patients report a worse outcome than an age-matched community-based sample, which indicates that none of the current treatment options are successful in restoring knee function from the patients' own point of view.

Return to sport

One of the major aims of treatment of ACL injury is to restore the knee and to return the athlete to pre-injury activity levels. This is also the main argument for surgical treatment. Reports of return to sports following ACL injury do not use a consistent definition and thus results should be interpreted with some caution. One frequent source of misinterpretation is that the definition of return to pre-injury activity is vague and includes one, and sometimes two, levels below the actual pre-injury activity level on a 10-graded scale.

Following ACLR, there is a large variation in reports of return to sport, with rates from 65–88% being able to return to sport within the first year.^{115–118} In a meta-analysis of 392 patients, 72% (n = 281) had returned to their pre-injury activity level 2 years after ACLR.¹¹⁹

A report on elite soccer players showed a return to sports rate of 94% after ACLR.¹²⁰ The authors speculate that one of the main contributing facts might be the extraordinary care and rehabilitation provided by the team physiotherapist in these professional soccer clubs. In contrast, another publication showed that only 63% of NFL athletes returned to NFL game play at an average of 10.8 months after ACL surgery.¹²¹ A success rate around 70% is not excellent for a surgical treatment option but it should be noted that some of these athletes do not return to sports for reasons other than knee problems. However, the proportion of such individuals is not well described.

A systematic review examined 69 articles, reporting on 7556 participants.¹²² On average, 81% of people returned to any sport, 65% returned to their pre-injury level of sport and 55% returned to competitive-level sport after surgery. Symmetrical hopping performance and the contextual factors of younger-age males playing elite sport and having a positive psychological response favoured returning to the pre-injury level sport. Another systematic review found weak evidence in 16 articles suggesting that variables associated with return to sport included higher quadriceps strength, less effusion, less pain, greater tibial rotation, higher Marx Activity score, higher athletic confidence, higher preoperative knee self-efficacy, lower kinesiophobia and higher preoperative self-motivation.¹²³

In a recent review, Nagelli and Hewett advocate waiting 2 years before RTP after ACL reconstruction.¹²⁴ They presented evidence showing that at 2 years after ACLR, biological healing (absence of bone bruises, ACL graft maturation and sensory restoration) has occurred and functional recovery (biomechanical and neuromuscular control and quadriceps strength) of the knee has normalised or is not significantly different from baseline.¹²⁴ Therefore, delaying a return to high-level activity for high-risk athletes until 2 years after ACLR will restore knee joint homeostasis and significantly reduce the risk of subsequent injury. They do, however, acknowledge the potential deleterious impact on an athlete's career if he or she misses 2 years of their sport.

Among patients treated non-operatively, the return to sports rates vary even more than among those treated surgically. Scientific reports suggest a range from 19% to 82% for return to pre-injury activity.^{67, 125–127} While there is no firm evidence to support that return to pre-injury sports is more likely after ACL reconstruction than after non-surgical treatment, individuals active on a professional or subprofessional level often lack the possibility to wait and see and thus undergo surgery in most cases. Athletes who successfully return to sport after non-operative treatment could represent a group gaining functionally stable knees without fully restored stability at rest. Factors associated with success after non-surgical treatment need to be better explored.

There is some evidence that those returning to pre-injury sports after ACL injury may stop playing earlier than their non-injured counterparts.¹²⁷⁻¹²⁹ In the only study in which the reduction in sport participation can be related to a control group, Roos et al.¹²⁶ found that only 30% of those who had ACL injury were active after 3 years compared with 80% of controls and that after 7 years none of the elite injured players were active regardless of the type of treatment. In addition, previously injured athletes retire at a higher rate than athletes without previous ACL injury.¹³⁰ The reason for this may be that many of the athletes who return to sport experience significant knee problems such as instability, reduced ROM and/or pain.¹²⁵

Criteria for RTP after ACL reconstruction

As previously mentioned, strength deficits persist following ACLR.¹⁰⁰ Various authors have recommended the use of isokinetic testing as a possible objective marker for RTP. In a review of the various protocols used in isokinetic testing involving 39 studies, the variables that were most commonly used were concentric/concentric mode of contraction (31 studies), angular velocity of 60°/s (29 studies), three to five repetitions (24/39 studies), ROM of 0-90° (6 studies) and using gravity correction (9 studies).¹³¹ Eight studies reported strength limb symmetry index scores as part of their RTP criteria. The authors concluded that there was no standardised isokinetic protocol following ACLR and that isokinetic strength measures have not been validated as useful predictors of successful RTP.¹³¹

In the only prospective study that addressed neuromuscular control and coordination, poor neuromuscular control was a risk factor for ACL graft rupture.¹³² In this study of 56 athletes, transverse plane net moment impulse at the hip, dynamic frontal plane knee ROM, side-to-side differences in sagittal plane, knee moment at initial contact and deficits in postural stability were associated with a three times greater risk of ACL graft rupture.¹³²

A review of guidelines used for RTP identified three objective criteria used to allow release to sports activities.¹³³ The most common was lower extremity muscle strength, followed by lower limb symmetry, and knee examination parameters of range of knee motion and effusion. Twelve studies listed one criterion for release to sports, eight studies listed two criteria and one study recommended three criteria.

A recent study followed 158 male professional athletes who underwent an ACLR and returned to their previous professional level of sport.¹³⁴ Before players returned to sport they underwent a battery of discharge tests (isokinetic strength testing at 60°, 180° and 300°/s, a running T-test,¹³⁵ single-hop, triple-hop and triple crossover-hop tests). The criteria for each test was as follows:

- isokinetic test at 60, 180 and 300°/s: quadriceps deficit <10% at 60°/s
- single-hop: limb symmetry index >90%
- triple-hop: limb symmetry index >90%
- triple crossover-hop: limb symmetry index >90%
- on-field sports-specific rehabilitation: fully completed
- running T-test <11 s.¹³⁵

The athletes returned to their previous competitive level a mean of 229 days after surgery (range 116-513 days). Of these, 26 (16.5%) sustained an ACL graft rupture and 11 (7.0% of 158) sustained a contralateral rupture of their native ACL. Athletes who did not meet the specific discharge criteria before return to sport and were not fully discharged from ACLR rehabilitation by the surgeon had a four times greater risk of sustaining an ACL graft rupture (73%) compared with those who returned to their sport after passing the necessary criteria (27%). The study also looked at hamstring to quadriceps ratio and found that those who had a lower ratio at 60°/s had a greater risk of ACL graft rupture (HR 10.6 per 10% difference, 95% confidence interval of 10.2 to 11, $p = 0.005$). The authors concluded that two factors were associated with increased risk of ACL graft rupture: 1. not meeting all six of the discharge criteria before returning to team training; and 2. decreased hamstring to quadriceps ratio of the involved leg at 60°/s.¹³⁴

Re-injury rate

The rate of graft re-rupture after 10 years is approximately 6%. It appears that the risk of re-rupture or a graft failure is highest within the first 12 months after an ACL injury.¹³⁶ After the first 12 months, the overall increase in risk is divided fairly evenly between the reconstructed knee and the contralateral knee.¹³⁷

The aetiology of ACL graft failure is varied and often more than one cause exists. In general, the aetiology of ACL graft failure can be divided into five categories: 1. surgical technique; 2. trauma; 3. poor graft incorporation and healing (failure of ACL graft to undergo the ligamentisation process); 4. arthrofibrosis; 5. inadequate rehabilitation. The most common reasons for failure due to surgical technique are non-anatomic tunnel placement, failure to address associated ligamentous laxity or secondary restraints at the time of the primary reconstruction, graft impingement, inadequate strength of the primary ACL graft and abnormal graft tension (under or overtensioning).

Early failures (<3 months) are usually related to loss of fixation and infection. Midterm failures (3-12 months) are most frequently due to errors in surgical technique, aggressive physical therapy, premature return to full activity or unrecognised loss of secondary restraints. Late failures (>12 months) are usually related to trauma.

The risk of a contralateral ACL injury is similar to that of a rupture of the graft (i.e. about 6%).^{137, 138} When reviewing the literature, it appears to be clear that the risk of contralateral ACL injuries is greater than the risk of first-time ACL injuries.¹³⁷ The most prominent risk factor for a contralateral ACL injury is the return to a high level of activity. There are some indications that young age at the initial ACL injury and at the subsequent ACLR,¹³⁹ ACLR with patellar tendon autograft (compared to hamstring tendon autograft) and a narrow intercondylar notch can increase the risk for a contralateral ACL injury. Female gender does not seem to be a risk factor for a contralateral ACL injury. Altered biomechanics and altered neuromuscular function as a result of the initial ACL injury, affecting both the injured and the uninjured leg, most likely, further increase the risk for a contralateral ACL injury.

Data from non-surgically treated individuals are lacking. One prospective investigation of consecutively recruited ACL injured patients, initially treated conservatively without ACLR with later surgical intervention only when deemed necessary, where the patients were advised to avoid contact sports, found an annual incidence of contralateral ACL injuries of 0.4% over 15 years.¹²⁷

Revision surgery

If failure of the original ACL graft occurs, a revision procedure can be performed. Outcomes of revision surgery are predictably not as good as in primary reconstruction. In a meta-analysis including studies with 300 revision ACLRs and 413 primary ACLRs, patients who had had revision surgery reported inferior Lysholm Knee Scoring Scale scores, had inferior clinician-reported knee function as assessed with the objective IKDC classification and pivot shift test and more radiographic evidence of tibiofemoral osteoarthritis (50% versus 25%) compared with patients who had had primary surgery.¹⁴⁰

A previous review included 31 articles which documented rate of return to sport and showed that whereas 73% had good objective and satisfactory subjective results, 57% of patients did not return to the same level of sport activity, significantly inferior to that of a primary procedure.¹⁴¹

Osteoarthritis

A frequent consequence of ACL rupture is the increased risk of knee osteoarthritis (OA).¹⁴² Following ACL injury, rates of radiographic knee OA are typically reported to be approximately 50% at 10–20 years post-injury.^{142, 143} As ACL injury frequently occurs in adolescents and young adults, these individuals typically develop OA before they reach 40 years, which would not be expected in the absence of joint trauma. An ACL injury is therefore said to be responsible for *early-onset* knee OA.

Current evidence does not indicate that ACLR reduces the rate of OA development.^{144, 145} Indeed, higher rates of OA may be evident following surgical compared to non-surgical management of the ACL-deficient knee, which may be related to the return to high-impact cutting and pivoting sports activity, which non-operative treatment algorithms often advise against.^{146, 147}

An ACL rupture causes a significant disruption to the homeostasis of the knee. The considerable force required to rupture a healthy ACL is consistent with the frequent presence of subchondral BMLs,¹⁴⁸ articular cartilage damage¹⁴⁹ and meniscal tears.^{148, 150} The considerable mechanical impact to the articular cartilage and subchondral bone can disrupt the cartilage matrix and alter cell metabolism,^{151, 152} which can predispose to OA development even in the absence of altered mechanical loading.¹⁵¹ However, the altered mechanical environment following an ACL injury is thought to be a fundamental driver for post-traumatic OA development.^{153, 154}

The instability and altered kinematics following ACL injury are theorised to shift load bearing to areas of cartilage unaccustomed to such load, potentially contributing to the early onset of OA.¹⁵³ Perhaps altered kinematics in the setting of already damaged cartilage is most deleterious following ACL injury, as cartilage response to load is dependent on the health of the cartilage.¹⁵³ Abnormal knee kinematics persist despite ACLR, which might explain why rates of OA are not reduced with surgical stabilisation.¹⁵⁵ Dynamic knee joint loading and OA development may also be affected by prolonged muscle weakness, in particular quadriceps weakness, which can persist for years after ACL injury.¹⁵⁶

Inherent biomechanical characteristics preceding ACL injury may also be implicated in the development of post-traumatic OA. Specifically, high external knee abduction loads during a jump-landing task can predict ACL injury in young female athletes,¹⁵⁷ while increased external knee abduction moments during gait have been associated with lateral tibiofemoral OA following ACLR.¹⁵⁸

The unique biomechanical characteristics following ACL injury, including high knee abduction loads, may contribute to the distinct patterns of post-traumatic disease typically observed.¹⁵⁹ Pre-injury kinematic and kinetic patterns are likely to persist following ACL injury, irrespective of surgical reconstruction, although targeted neuromuscular training may help to decrease knee abduction loads.¹⁶⁰

Importantly, post-traumatic knee OA displays distinct patterns of disease compared to a typical non-traumatic OA presentation. Post-traumatic OA affects the medial and lateral tibiofemoral compartment equally, likely reflecting the local factors unique to ACL injury and reconstruction.^{158, 159}

Patellofemoral OA is also particularly prevalent at all time points following ACLR regardless of autograft type^{161, 162} and early signs of patellofemoral deterioration are evidence as early as 1 year post-surgery.¹⁶³ Altered biomechanics, including tibial rotations and knee flexion, may play a role in patellofemoral OA following ACLR^{164, 165} and lower limb function or the presence of patellofemoral cartilage lesions can predict the presence of poorer prognosis following ACLR.¹⁶⁶ Post-traumatic OA should therefore be considered a distinct entity and clinicians should pay particular attention to the status of the patellofemoral joint following ACLR.

The increased risk of OA development following ACL injury and reconstruction appears to be driven by factors such as concomitant trauma to other knee structures, altered kinematics and loading environment, and muscle weakness. However, systemic factors that increase the risk of incident non-traumatic OA, such as older age, female sex and higher body mass index (BMI), also appear to accentuate the development of post-traumatic OA. The interaction of

local and systemic risk factors likely creates an imbalance of biochemical markers, leading to cartilage degradation and ultimately OA. This is summarised in Figure 35.14.

Sex difference

In light of the increased prevalence of ACL injuries in female athletes discussed previously, attention has been paid to possible differences in outcome after ACLR between males and females. The majority of studies show increased post-surgical laxity in females, but no difference in graft failure, activity level, or subjective or functional assessment.^{115, 167-171}

The most recent review included 13 studies.¹⁷² Meta-analysis revealed no difference in graft failure risk (eight studies), contralateral ACL rupture risk (three studies) or postoperative knee laxity on physical examination (six studies). There was no evidence of a clinically important difference in patient-reported outcomes according to sex.

However, one study of several thousand patients treated with ACLR from the Swedish ACL registry showed

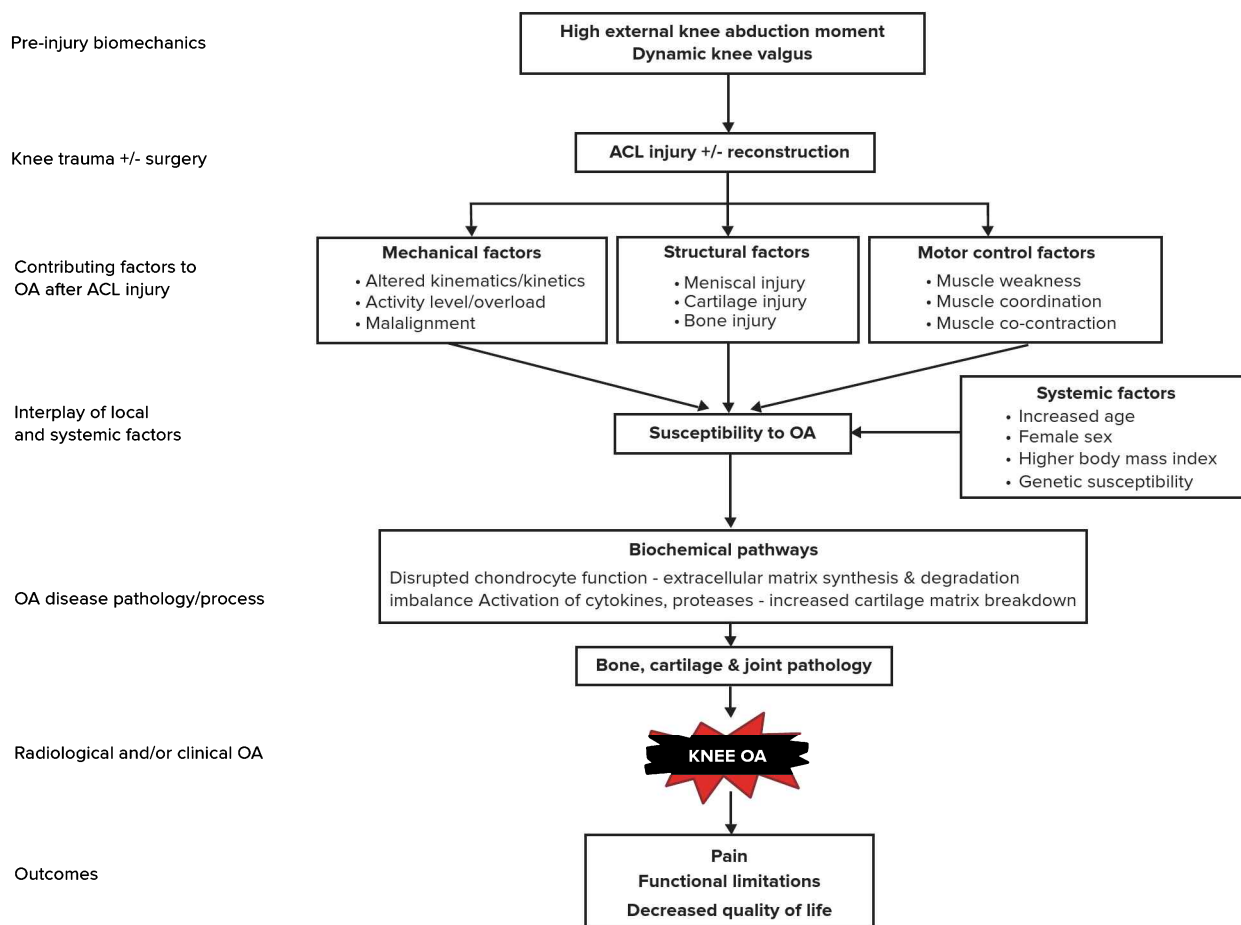


Figure 35.14 Development of osteoarthritis following ACL injury

COURTESY DR ADAM CULVENOR

significant gender differences in the self-reported outcome 1 and 2 years after surgery.¹¹² The authors reported that female patients showed worse outcomes than male patients before surgery and at 1 and 2 years after ACLR, and suggested that possible sex differences should be analysed in future studies on evaluation after ACL injury/reconstruction.

Age

Younger athletes have been shown to have a higher chance of graft rupture than older athletes.¹⁷³ A 10-year difference in age was associated with a 2.6 times greater chance of ACL graft rupture. Among 1820 patients after primary ACLR, younger athletes had a higher activity level and a higher incidence of ACL re-injury.¹⁷⁴ Based on data from the Swedish National Knee Ligament Register, the combination of being young and playing soccer is a substantial risk factor for revision surgery.¹⁷⁵

Results after ACL reconstruction in patients older than 40 years and comparison with younger populations were recently the subject of a systematic review.¹⁷⁶ Twelve studies with a total of 452 patients were included. The mean patient age was 47.8 years (range, 40–66 years) with a mean follow-up of 53.3 months (minimum, 24 months). Lysholm scores improved from 53.9 to 90.5 in the 11 operative studies. IKDC scores of A or B were found in 81%. Tegner activity scores averaged 4.7 pre-injury, fell to 2.9 preoperatively and returned to 4.7 postoperatively. The reported failure rate was 2.3%. There were few complications and failure rate was similar in younger patients. The data confirm that ACLR can be recommended to patients older than 40 years who wish to maintain an active lifestyle or have symptomatic instability with daily activities.

Partial ACL tear

Approximately 5–15% of ACL injuries are thought to be partial tears.¹⁷⁷ There is no consensus on the definition of a partial ACL tear. The presence of an intra-substance incomplete tear of the ACL on MRI (Fig. 35.15) or at arthroscopy is generally regarded as being the gold standard diagnosis. Rather than an anatomical diagnosis, a functional assessment of stability is more practical from a treatment viewpoint.

If clinical instability is present and the pivot shift is positive, then ACL instability is diagnosed. Most would consider a 'partial' tear to be associated with a negative pivot shift test. Controversy centres around whether a complete tear of one of the two bundles (anteromedial and posterolateral) should be considered under the definition of a partial tear and whether some asymmetrical laxity on Lachman's test albeit with a definite end point should be included.¹⁷⁸ The partial tear is often described on MRI reports as a percentage of ACL remaining.

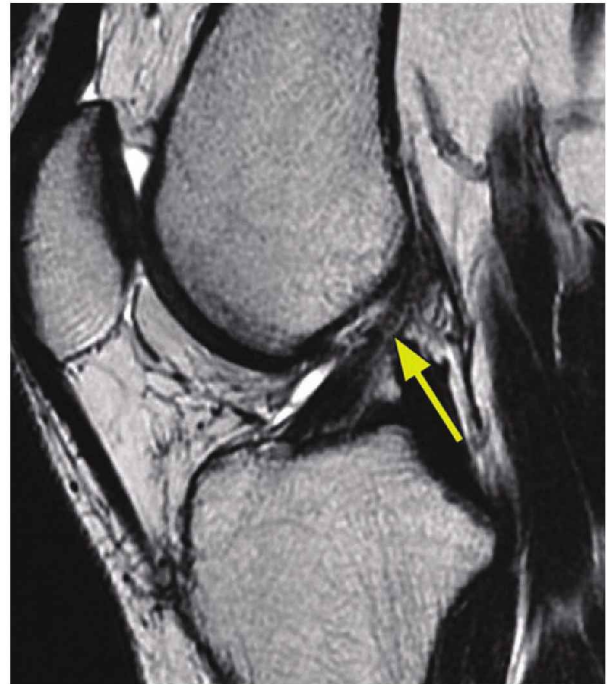


Figure 35.15 Sagittal T2 image showing partial ACL rupture characterised by increased T2 signal (yellow arrow) and some fibre disruption within its proximal aspect. The normal angle and some continuous fibres distinguish this partial tear from a full-thickness rupture

The management of partial ACL tears depends on the degree of instability. A clinically and functionally stable partial tear can be treated non-surgically with the option of later surgery if needed. Unstable lesions especially in those who wish to play high-intensity sports can be treated surgically either with a full knee reconstruction or augmentation of the remaining ligament. Degenerative changes were present in 15% of patients 8 years after partial ACL tear managed conservatively.¹⁷⁹

Prevention of ACL injuries

Prevention of ACL injuries is thoroughly discussed in Chapter 12. Risk factors and prevention programs including the FIFA-11 are described in detail.

ACL rupture in children with open physes

Rupture of the ACL in children with open physes presents an additional challenge to the treating practitioner. This topic is covered extensively in Chapter 44.

POSTERIOR CRUCIATE LIGAMENT INJURIES

The PCL is composed of two bundles, the larger anterolateral (AL) bundle and the smaller posteromedial (PM) bundle, which are most readily identified at their femoral locations.

However, at their tibial attachment to the PCL facet, the bundles are more compact and difficult to separate. Historically, the two bundles of the PCL were believed to function independently, with the ALB primarily functioning in flexion and the PMB in extension. However, in light of recent biomechanical investigations, a more synergistic and co-dominant relationship between the two bundles has been validated. Ahmad et al. investigated the spatial orientation of the PCL bundles and reported that changes in the orientation of each bundle during knee flexion and extension prevented either bundle from exercising complete dominance in the restraint of posterior tibial motion.¹⁸⁰

The PCL has historically been considered a primary restraint to posterior tibial translation and more recent studies have also identified it as a secondary restraint to rotation, particularly between 90° and 120° of flexion.¹⁸¹ Complete sectioning of the PCL has reportedly led to a significant increase in posterior tibial translation at 0° to 120° of flexion and an increase in internal rotation at 90° to 120° of flexion. The magnitude of abnormal translation and rotation appears to depend on whether one or both of the bundles are injured. An isolated tear of either PCL bundle results in a minimal clinically important increase in posterior tibial translation. Generally, the literature supports that the PCL has a more expansive role in providing rotational stability than previously thought, and it is important to assess internal and external rotation stability when considering PCL injury.¹⁸¹

PCL injury typically presents concurrently with other knee injuries, including ACL, MCL or posterolateral corner (PLC) injury or, less commonly, in isolation. Recent studies have helped to clarify and elucidate the prevalence of concurrent ligament injuries with a complete (grade III) PCL tear.¹⁸¹ A majority of grade III PCL tears are associated with multi-ligament knee injuries, with one study reporting that 79% of multi-ligament knee injuries involved the PCL in a trauma setting.¹⁸² Furthermore, 46%, 31% and 62% of PCL injuries have been reported to be concomitant with ACL, MCL and PLC knee injury in a trauma setting, respectively.¹⁸³

PCL injuries are often associated with meniscal and chondral injury. The incidence of associated meniscal tears varies from 16% to 28%. Longitudinal tears of the anterior horn of the lateral meniscus are the most common location. There is also a high incidence of radial tears in the middle or posterior lateral meniscus.¹⁸⁴ The incidence of significant chondral damage with isolated PCL injury was not thought to be as high as with ACL injury, but one study showed chondral damage in 52% of those with PCL tears, with lesions of grade III or more found in 16%.¹⁸⁴

Clinical features

The mechanism of PCL injury is usually a direct blow to the anterior tibia with the knee in a flexed position (i.e. dash-board injury). This can be from contact with an opponent, equipment or falling onto the hyperflexed

knee. Hyperextension may also result in an injury to the PCL and posterior capsule.

PRACTICE PEARL

Whereas a patient with an ACL or MCL injury often describes the feel of a distinct 'pop' or 'tear', PCL tears typically have vague symptoms such as unsteadiness or discomfort.

A patient with an acute isolated PCL injury may have a mild-to-moderate effusion, pain in the posterior aspect of the knee or pain with kneeling. In a case of subacute or chronic PCL injury, the patient may describe vague anterior knee pain, pain with deceleration and descending inclines or stairs, or pain with running at full stride.

On examination, there is usually minimal swelling as the PCL is an extrasynovial structure. The posterior drawer test (Fig. 35.2o) is the most sensitive test for PCL deficiency. This is performed in neutral, internal and external rotation. A posterior sag of the tibia (Fig. 35.2n), and pain and laxity on a reverse Lachman's test, may be present. PCL rupture is particularly disabling for downhill skiers, who rely on this ligament for stability in the tucked up position adopted in racing.

PCL tears are graded I, II and III on the position of the medial tibial plateau relative to the medial femoral condyle at 90° of knee flexion (the posterior drawer position). The tibia normally lies approximately 1 cm anterior to the femoral condyles in the resting position. In grade I injuries, the tibia continues to lie anteriorly to the femoral condyles but is slightly diminished (0-5 mm laxity). In grade II injuries, the tibia is flush with the condyles (5-10 mm laxity). When the tibia no longer has a medial step and can be pushed beyond the medial femoral condyle (>10 mm laxity), it is classified as a grade III injury.

It is important to distinguish between isolated PCL injury and a combined PCL and posterolateral corner injury. In isolated PCL tears, there is a decrease in tibial translation in internal rotation due primarily to the influence of the MCL.¹⁸⁵

Plain radiographs are advocated when one is examining the presence of avulsion fracture fragments, Segond fractures, fibular head avulsions and lateral joint space widening. Any perceptible degree of posterior tibial sag should be documented. Stress radiographs provide reproducible objective assessment of the degree of posterior tibial translation between the injured and normal contralateral knee.¹⁸⁶ It has been reported that partial PCL tears result in <8 mm of increased posterior tibial translation, isolated complete PCL tears in 8-12 mm and combined complete PCL tears (usually with a PLC injury) in >12 mm.¹⁸⁷

MRI has a high predictive accuracy in the diagnosis of the acute PCL injury,^{188, 189} but a lesser accuracy in chronic injuries.¹⁹⁰ The normal appearance of a PCL is a

well-defined continuous band of low signal intensity in all pulse sequences with a maximum anteroposterior diameter of 6 mm when measured on sagittal T2-weighted images. Conversely, a torn PCL (Fig. 35.16) has been reported to have an abnormally large (>7 mm) anteroposterior diameter.¹⁹¹

Bone bruises are seen in over 80 % of acute grade II and III PCL injuries on MRI. In contrast to the bone bruises associated with ACL tears, which commonly occur on the posterior tibia and lateral femoral condyle, the location of bone bruises observed with PCL injuries is less predictable. Bone bruises in the medial compartment were significantly more often associated with a PCL-PLC combined injury, while lateral compartment bone bruises were significantly more often associated with a concomitant PCL-MCL injury.¹⁹²

Treatment

Grade I, II and most grade III PCL injuries can generally be managed non-surgically with a comprehensive rehabilitation program. A suggested program emphasising intensive quadriceps exercises is shown in Table 35.11. More severe injuries (grade III) should be immobilised in extension for the first 2 weeks.

The PCL has been reported to have an intrinsic healing ability after injury, although this healing may occur in a lax or attenuated position. Therefore, in cases of acute isolated PCL injury, non-operative management has been recommended. However, acute multi-ligament knee injuries with a concomitant or chronic PCL tear are believed to be best treated by surgery.¹⁸⁰ The use of a brace (e.g. the Jack or Rebound PCL brace) that applies a constant or dynamic

anterior force to counteract the posterior sag of the tibia, may help to increase healing after PCL injury by reducing the ligament to a more physiological position.¹⁹³

Patients with isolated PCL tears have a good functional result despite ongoing laxity after an appropriate rehabilitation program. Regardless of the amount of laxity, half of the patients in one large study returned to sport at the same or higher level, one-third at a lower level and one-sixth did not return to the same sport.¹⁹⁴ Other studies have shown that knee instability did not correlate with return to sport or knee satisfaction.¹⁹⁵⁻¹⁹⁷

Although favourable non-operative results have been observed, the long-term consequences of isolated PCL deficiency remain unknown.¹⁹⁸ Boynton and Tietjens have observed deterioration at extended follow-up of non-operative treatment despite good early results.¹⁹⁹ In 38 patients with isolated tears with a mean follow-up of 13.4 years, eight patients had subsequent meniscal injuries and surgery. Of the remaining 30 patients with normal menisci, over 80% had occasional pain and over 50% had occasional swelling. With time, an increase in articular cartilage degeneration was seen on plain radiographs. Investigators who evaluated isolated PCL tears that were treated with non-operative rehabilitation programs reported radiographic evidence of arthritic changes in 23% of patients at 7-year follow-up²⁰⁰ and 41% at 14-year follow-up.²⁰¹ However, only a small percentage of the patients in the long-term follow-up had moderate or severe osteoarthritis (11%), and the majority of patients had good strength and subjective outcome scores, although stress radiographs were not reported.¹⁸⁰

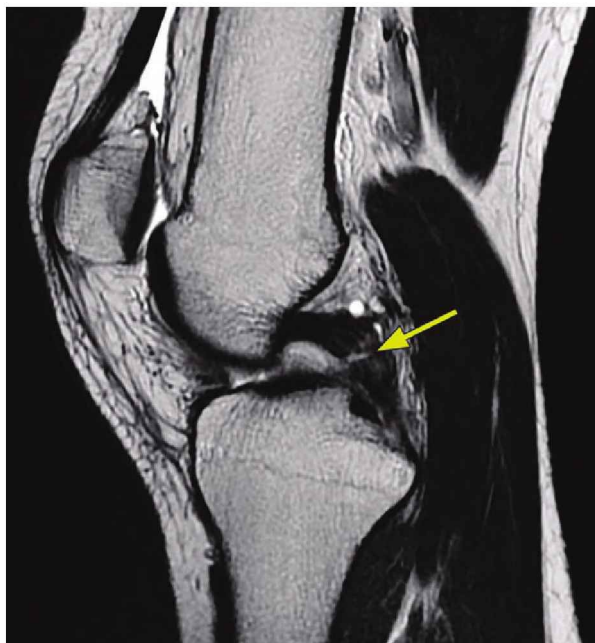


Figure 35.16 Sagittal T2 MRI showing a defect with associated oedema (arrow) in the midsubstance of the PCL

Indications for surgical repair

Operative management is reserved for acute or chronic isolated grade III PCL injuries with symptoms of pain or instability which have failed an adequate rehabilitation program treatment, or PCL insufficiency in the setting of a multi-ligamentous knee injury.¹⁹⁸ Complete PCL tears, as indicated by PCL stress radiographs with an increased anteroposterior laxity measurement of >8 mm, combined with repairable meniscal body or root tears are a possible indication for PCL reconstruction.¹⁸⁰

Similar to ACL reconstruction, grafts can be either autograft (hamstring tendon or bone-patellar tendon-bone) or allograft, single or double bundle. Single-bundle PCL reconstruction is typically indicated for symptomatic chronic PCL injuries, either in isolation or combined with other knee ligamentous injury, and for acute, multi-ligament PCL injuries. Recent efforts have focused on an anatomic single-bundle reconstruction using arthroscopic and radiographic reference points instead of the historical non-anatomic 'isometric' reconstruction that has been reported to result in initial joint overconstraint and increased laxity over time.

Controversy exists regarding the use of transtibial versus tibial inlay techniques of PCL reconstruction. In the transtibial

Table 35.11 Rehabilitation of a PCL tear (see Figs 35.6 and 35.7)

Phase	Goal of phase	Time post-injury	Physiotherapy treatment	Exercise program	Functional/sport-related activity
Phase 1	PWB–FWB Eliminate swelling 0–100° ROM 4+/5 quadriceps strength 5/5 hamstring strength	0–2 weeks	Jack or Rebound PCL brace Cryotherapy Electrotherapy Compression Manual therapy Gait re-education Patient education	Gentle flexion ROM Extension ROM to 0° Quadriceps/VMO setting Supported (bilateral) calf raises Hip abduction and extension Hamstring pulleys/rubbers Gait drills	Nil
Phase 2	No swelling Full ROM 4+/5 quadriceps strength 5/5 hamstring strength	2–4 weeks	Cryotherapy Electrotherapy Compression Manual therapy Gait re-education Exercise modification	ROM drills Quadriceps/VMO setting Mini-squats and lunges Leg press (double, then single-leg) Step-ups Bridges (double, then single-leg) Hip abduction and extension with rubber tubing Single-leg calf raises Gait re-education drills Balance and proprioceptive drills (single-leg)	Walking Exercise bike
Phase 3	Full ROM Full strength and power Return to jogging, running and agility Return to restricted sport-specific drills	4–6 weeks	Manual therapy Exercise/activity modification and supervision	As above—increase difficulty, repetitions and weight where appropriate Jump and land drills Agility drills	Straight-line jogging Swimming (light kick) Road bike Straight-line running Progressing to sport-specific running and agility (progressively sequenced) (e.g. running forwards, sideways, backwards, sprinting, jumping, hopping, changing directions, kicking)
Phase 4	Return to sport	6–10 weeks	As above	High-level sport-specific strengthening as required	Progressive return to sport (e.g. restricted training, unrestricted training, match play, competitive match play)

FWB = full weight-bearing; PWB = partial weight-bearing; ROM = range of motion; VMO = vastus medialis obliquus

technique, the tibial and femoral tunnels are drilled and the graft must make an acute turn as it surfaces from the tibial tunnel and changes direction before entering the knee joint. Subsequent graft abrasion and attenuation may result in graft rupture or laxity.²⁰² This sharp turn has been implicated in the residual posterior knee laxity observed clinically after transtibial PCL reconstruction. The tibial inlay was developed as an alternative to the transtibial technique to avoid this problem. In tibial inlay reconstruction, the graft is fixed directly to the PCL's native insertion site on the tibia.¹⁹⁸

Double-bundle PCL reconstructions have evolved as an alternative to single-bundle reconstructions with the same indications for surgery. As reported by biomechanical studies, the AL and PM bundles perform in a co-dominant manner and these roles theoretically would not be restored by a single-bundle PCL reconstruction.²⁰³ Therefore, an anatomic double-bundle PCL reconstruction may be able to more closely restore native kinematics than the single-bundle technique.

Results of surgical reconstruction are generally good,^{204, 205} but one review²⁰⁶ suggested that the rate of graft failure was 11.6%. In the isolated PCL studies, 50–82% of patients were able to return to pre-injury activity level. The most commonly reported complications after PCL reconstructions are residual posterior laxity, usually defined as more than 4 mm of increased posterior translation on PCL stress radiographs, and flexion loss due to prolonged immobilisation of the knee in extension.

Since PCL graft healing times have been reported to be almost double the time of ACL graft healing, it has been suggested that PCL reconstruction patients should be kept non weight-bearing for 6 weeks. Patients are placed in an immobiliser brace in extension after surgery for 3 days before transitioning to a dynamic or static anterior drawer knee brace. The Jack Brace and the Rebound® PCL brace which apply increasing force as a function of flexion angle, are the only currently available dynamic braces and it has been recommended that the brace of choice should be worn at all times for up to a minimum of 24 weeks postoperatively.²⁰⁷

LATERAL COLLATERAL LIGAMENT TEARS

Lateral (or fibular) collateral ligament (LCL) tears are much less common than MCL tears. In contrast to the common pattern of isolated MCL injury, an injury to the lateral side of the knee usually requires higher magnitude of force and thereby affects additional structures in the area. Isolated injuries to the LCL involve forces of lower magnitude, resulting in lower-grade damage to the ligament. Isolated high-grade injuries to the LCL are therefore fairly rare.²⁰⁸

Posterolateral corner injuries

Posterolateral corner injuries are being increasingly recognised. In a consecutive series of 187 patients with

an acute knee injury and a complete ligament tear found on MRI, the incidence of grade III PLC injuries was 16%.²⁰⁹

The posterolateral corner (PLC) of the knee was once referred to as 'the dark side of the knee' due to the limited understanding of the structures, biomechanics and possible treatment options.²¹⁰ A number of studies in recent years have led to a heightened understanding of the PLC and biomechanically validated reconstruction techniques. Posterolateral corner injuries are commonly associated with ACL or PCL tears. The LCL, popliteofibular ligament, arcuate ligament, oblique popliteal ligament, fabellofibular ligament, posterolateral capsule, popliteus tendon, biceps femoris tendon, lateral head of the gastrocnemius, lateral meniscus and the iliotibial band may all be involved in the injury pattern to some degree.

The three major static stabilisers of the PLC are the LCL, the popliteus tendon (PLT) and the popliteofibular ligament (PFL).²¹¹ The PLC structures provide the primary restraint to varus forces of the knee as well as posterolateral rotation of the tibia relative to the femur in cases of cruciate deficient knees. These structures are also important secondary stabilisers to anterior and posterior tibial translation.

Mechanisms of injury usually cause force at multiple vectors of varying magnitude—with varus, extension, rotation and translation all playing a role. A common mechanism of injury to the PLC is a direct blow to the anteromedial knee. However, hyperextension and non-contact varus stress injuries can also damage the PLC. Most often, PLC tears are associated with ACL and/or PCL injuries, justifying close examination for a PLC injury for every cruciate ligament tear.²¹⁰

Frequently reported symptoms include pain, perceived side-to-side instability near extension, increased difficulty walking on uneven ground or up and down stairs, bruising and swelling. This instability and difficulty walking can present as a varus thrust gait seen during the initiation of the stance phase.²¹² It is not uncommon for the patient to complain of paraesthesia of the common peroneal nerve distribution or foot drop. Common peroneal nerve injury has been reported to occur in up to one-third of PLC injuries.²¹³

The varus stress (Fig. 35.2g) test should be performed in both full extension and at 20–30° of flexion. Lateral compartment gapping in comparison to the contralateral side with the knee flexed to 30° indicates an injury to the LCL and potentially to the secondary stabilisers of the PLC. If stability is restored when tested in full extension, an isolated injury to the LCL is presumed. However, if the varus instability persists in full extension, a combined LCL, PLC and cruciate ligament injury is assumed.²¹⁰

A routine radiographic work-up with standing anteroposterior (AP), lateral and axial views should be performed to rule out the presence of fractures. A standing

long-leg AP view should be obtained in chronic cases because the limb alignment should be corrected using an osteotomy prior to or at the same time as a reconstruction procedure. Additionally, varus and PCL stress radiographs can be used to obtain objective quantification of the amount of lateral compartment varus gapping and a combined PLC and PCL injury, respectively.²¹⁴

MRI allows identification of concurrent lesions such as meniscus tears, cartilage lesions and occult fractures. It has been shown to have 90% sensitivity and specificity for IT band, biceps tendon, LCL and PLT injury. The only PLC structure with lower diagnostic accuracy values was the popliteofibular ligament, with 68.8% sensitivity and 66.7% specificity.²⁰⁹ For optimal MRI diagnostic accuracy for PLC injuries, an imaging sequence using 2 mm slices in a coronal oblique plane following the obliquity of the PLT should be employed. Bone bruise patterns are found in 81% of all PLC injuries, usually on the anteromedial femoral condyle. Together, these imaging techniques are excellent tools to augment the diagnosis of PLC injury.²¹⁴

Non-surgical management

Good results have been reported for non-surgical treatment for grade I and II injuries with minimal radiographic changes at 8 year follow-up following the use of an early mobilisation protocol.²¹⁵ However, poor functional outcomes for non-operatively treated grade III PLC injuries with persistent instability and degenerative changes have been reported.²¹⁶ Increased forces on the PCL and ACL reconstruction grafts have been reported if concurrent PLC injuries are not addressed.²¹⁷

Surgical management

Surgical treatment of PLC injuries is the treatment of choice for patients with isolated grade III PLC injuries, combined PLC injuries and failed non-operative treatment.²¹⁴ Acute surgical treatment (<3 weeks) results in improved outcomes and can avoid the need for an additional procedure for limb alignment correction that may be necessary in chronic cases. Patients treated acutely may undergo repair or reconstruction procedures. Primary repairs of LCL and PLTs avulsions, without midsubstance injury, may be performed within 2–3 weeks after the injury. After that point, the tissue becomes retracted and scars down, making it nearly impossible to reattach the injured structures to their native anatomic locations. Improved outcomes have been demonstrated with primary repair of avulsed ligament structures and augmentation of midsubstance ligament injuries.

Several reconstruction techniques have been described and can be broadly categorised as biceps tendon transfer, fibular-based sling or anatomic-based reconstruction of the LCL, PLT and popliteofibular ligament.²¹⁸ Outcomes following reconstruction have been shown to be

significantly better than those following repair.^{219, 220} A recent systematic review found the repair of acute grade III PLC injuries with staged cruciate reconstruction was associated with a 38% failure rate, whereas a more robust reconstruction-focused approach for PLC injuries with concurrent reconstruction of cruciate injuries resulted in an overall mean 9% failure rate.²¹⁸

Surgical reconstruction of a chronic LCL tear is difficult. Surgical techniques include variations of fibular slings, capsular shifts and anatomic-based techniques (fibular tunnel and tibial tunnel). A recent systematic review included 15 studies with a total of 456 patients of which 59% had combined PCL injuries, 23% had combined ACL injuries, 6% had combined ACL and PCL injuries and 12% had isolated PLC injuries.²²¹ Cruciate ligament tears were treated simultaneously with PLC reconstruction in the majority of the 15 studies. Surgical management of these chronic PLC injuries had a 90% success rate and a 10% failure rate according to the authors' objective assessment of outcomes.²²¹

In chronic injuries (greater than 6 weeks from injury), lower extremity alignment should be evaluated and corrected with an osteotomy prior to ligament reconstruction because failure to address malalignment can lead to increased stress and stretching of the reconstruction grafts and failure.²²² A varus knee with lateral and/or posterolateral instability is associated with worse results.

ARTICULAR CARTILAGE DAMAGE

Chondral defects are seen in 3.4–62% of knee arthroscopies, while full-thickness focal lesions with an area of at least 1–2 cm² are seen in 4.2–6.2% of all arthroscopies in patients younger than 40 years of age.²²³ The incidence is higher in athletes.

Epidemiology

A systematic review included 11 studies containing 931 athletes of which 40% were professionals (US National Basketball Association and National Football League).²²³ The overall prevalence of full-thickness focal chondral defects in athletes was 36%. Fourteen per cent of athletes were asymptomatic at the time of diagnosis. Patellofemoral defects (37%) were more common than femoral condyle (35%) and tibial plateau defects (25%). Medial condyle defects were more common than lateral (68% versus 32%) and patella defects were more common than trochlea (64% versus 36%). Meniscal tear (47%) was the most common concomitant knee pathological finding, followed by ACL tear (30%) and then medial collateral ligament or lateral collateral ligament tear (14%).

Articular cartilage damage may occur as an isolated condition in which chondral or subchondral damage is the primary pathology or in association with other injuries, such as ligamentous instability resulting from MCL, ACL or PCL injuries or patellar dislocations.

Table 35.12 Outerbridge classification of chondral defects

1. Softening
2. <1 cm (<0.4 in.) partial thickness lesion
3. >1 cm (>0.4 in.) defect, deeper
4. Subchondral bone exposed

Table 35.13 International Cartilage Repair Society (ICRS) classification of chondral defects

1. Superficial lesions
 - A. Soft indentation
 - B. Superficial fissures or cracks
2. Lesions <50% cartilage depth
3. A. Lesions >50% depth
 - B. Down to calcified layer
 - C. Down to but not through subchondral bone
 - D. Blisters
4. Full thickness into subchondral bone

Articular cartilage damage may also be seen in association with meniscal injury and patellar dislocation.

Quantifying chondral injury

Chondral injury is graded according to the Outerbridge classification and more recently the International Cartilage Repair Society (ICRS) grading system (Tables 35.12 and 35.13). Articular cartilage damage varies from gross, macroscopically evident defects in which the underlying bone is exposed (grade IV), to microscopic damage that appears normal on arthroscopy but is soft when probed (grade I).

Articular cartilage damage in the knee has both short-term and long-term effects. In the short term, it causes recurrent pain and swelling. In the longer term, it accelerates the development of osteoarthritis. Various methods have been used to encourage healing of articular cartilage defects. These include microfracture (piercing the subchondral bone with an 'ice pick' to recruit pluripotent stem cells from the marrow), mosaic plasty such as Osteoarticular Transfer System Surgery or OATS (osteochondral plugs are taken from the trochlea margin and implanted within the injured area) and autologous chondrocyte implantation (ACI), where cultured chondrocytes (harvested from the patient and cultured in the laboratory) are re-implanted to the chondral defect. Gene therapy and bone morphogenetic proteins are currently in the experimental stage.

Management

There is currently considerable debate as to the efficacy of the various treatments and as yet no consensus on optimal treatment has been reached. Although short-term reduction

of symptoms has been shown with these treatments, long-term reduction of arthritic disability has not been shown. Successful results also vary according to size and location of the articular lesion and between treatments. As yet, no method of treatment has been able to reproduce true hyaline cartilage with its complex layered structure.

A systematic review to determine which surgical technique/s have improved outcomes and enabled athletes to return to their pre-injury level of sports, and which patient and defect factors significantly affect outcomes after cartilage repair or restoration identified 11 studies for inclusion (658 subjects).²²⁴ Only one RCT was identified. All other studies were prospective cohorts, case-control studies, or case series reporting results after microfracture, ACI or OATS. Better clinical outcomes were observed after ACI and OATS versus microfracture. Results after microfracture tended to deteriorate with time. The overall rate of return to pre-injury level of sports was 66%. The timing of return to the pre-injury level of sports was fastest after OATS and slowest after ACI. Defect size of less than 2 cm², preoperative duration of symptoms of less than 18 months, no prior surgical treatment, younger patient age and higher pre-injury and post-surgical level of sports all correlated with improved outcomes after cartilage repair, especially ACI. Results after microfracture were worse with larger defects. The rate of return to sports was generally lower after microfracture versus ACI or OATS and if a patient was able to return to sports, performance was diminished as well.

Another review surveyed the literature regarding factors used in determining a course of surgical treatment for symptomatic cartilage lesions of the knee to determine which factors affect treatment outcomes and should be incorporated in the treatment algorithm.²²⁵ Twenty-seven studies examining 1450 human (1416 in vivo; 34 cadaveric) and 90 animal subjects met inclusion criteria. Female sex and higher BMI significantly predicted cartilage loss rates and recovery after microfracture and autologous matrix-induced chondrogenesis. Defect size and location significantly predicted treatment outcomes. Sizes >2–4 cm demonstrated worse outcomes after microfracture treatment. Defect size did not consistently affect autologous chondrocyte implantation or osteochondral autograft transplantation outcomes. Intra-articular lesion location was related to intralesional subchondral bone contact and microfracture outcome. Corrected patellofemoral and tibiofemoral alignment improved clinical outcome when realignment procedures were done concurrently with cartilage repair.

The authors concluded that the choice of the appropriate repair technique for focal knee cartilage defects is multifactorial and suggested a treatment algorithm which should consider frequently used factors such as defect size, location, knee alignment and patient demand. However, patient sex and BMI could also be considered. Patient age was not significantly associated with clinical outcome.

An effective method of promoting scar tissue formation in damaged articular cartilage is by continuous passive motion. Continuous passive motion has been shown to stimulate formation of hyaline-like fibrocartilage in the chondral defect, especially in the immediate postoperative period. This should be supplemented by low load, non weight-bearing exercise such as swimming and cycling. Following articular cartilage injury, the athlete may have to modify his or her training to reduce the amount of weight-bearing activity and substitute activities such as swimming and cycling.

When an injury (e.g. patellar dislocation, ACL or MCL tear) requires a lengthy period of partial or non weight-bearing, particular attention must be paid to preserving the integrity of the articular cartilage. This is done with continuous passive motion, a hydrotherapy program, swimming or cycling.

Other methods of reducing stress on the damaged articular cartilage include correction of biomechanical abnormalities, attention to ensure symmetry of gait and the use of a brace to control any instability. Pool running, mini-trampoline and the AlterG® Anti-Gravity Treadmill® may be used in the early stages of running and agility work to reduce load bearing. Proprioceptive exercises and strength exercises are also important.

ACUTE PATELLAR TRAUMA

Acute trauma to the patella (e.g. from a hockey stick or from a fall onto the kneecap) can cause a range of injuries from fracture of the patella to osteochondral damage of the patellofemoral joint with persisting patellofemoral joint pain. In some athletes, the pain settles without any long-term sequelae.

If there is suspicion of fracture, a plain radiograph should be obtained. It is important to be able to differentiate between a fracture of the patella and a bipartite patella which is a benign finding present in 1-2% of the population. A skyline view of the patella should be performed in addition to normal views.

If there is no evidence of fracture, the patient can be assumed to be suffering acute patellofemoral pain. This can be a difficult condition to treat. Treatment consists of nonsteroidal anti-inflammatory drugs (NSAIDs), local electrotherapy (e.g. interferential stimulation, TENS) and avoidance of aggravating activities such as squatting or walking down stairs. Taping of the patella may alter the mechanics of patellar tracking and therefore reduce the irritation and pain (Chapter 36). If taping provides symptom relief, rehabilitation and altered loading by strengthened quadriceps could be beneficial.

Fracture of the patella

Patellar fractures can occur either by direct trauma, in which case the surrounding retinaculum can be intact, or by

indirect injury from quadriceps contraction, in which case the retinaculum and the vastus muscles are usually torn.

Fractures should be described by fracture orientation and position (e.g. transverse, inferior pole), degree of comminution and displacement (especially affecting the articular surface). Vertical fractures are possible but rare. A well-corticated fragment in the superolateral quadrant is more than likely part of a bipartite patella and not an acute fracture. Avulsion fractures are most commonly located at the inferior pole of the patella and often encountered in the paediatric population as patellar sleeve fractures. In the adult population, avulsion fractures occur at the origin of the patellar tendon and often cause disruption of the extensor mechanism.

The goals of treatment are a functional extensor mechanism, articular congruity and full, painless motion of the knee. Undisplaced fractures of the patella with normal function of the extensor mechanism can be managed non-surgically, initially with an extension splint. Over the next weeks as the fracture unites, the range of flexion can be gradually increased and the quadriceps strengthened in the inner range.

Transverse fractures are best treated with a tension-band technique (traditionally two axial Kirschner wires with a figure-of-eight wire anteriorly) which neutralises tension forces anteriorly and converts them into stabilising compressive forces at the articular surface. This technique is stronger than a simple cerclage wire in transverse fractures and can be combined with screw fixation in simple and in more comminuted fractures to increase stability.²²⁶

Small fragments may be excised or repaired with suture or screw fixation if possible. Every effort should be made to preserve and stabilise as much of the patella as possible to maintain the moment arm and avoid loss or poor outcomes which can occur when greater than 40% of the patella is removed. Removal of the entire patella may be contemplated for patients with massively comminuted fractures. Excision addresses the pain related to the fracture but decreases the mechanical advantage the patella provides to the extensor mechanism. The loss of quadriceps strength after total patellectomy can be as great as 49% and enough loss to preclude climbing stairs or getting out of a chair. Therefore, total patellectomy should be considered only after all techniques for salvage have been exhausted.²²⁶

Patella dislocation

Mean annual incidence of patellar dislocation varies with age group: it is between 5.8 and 7.0 per 100 000 person-years in the general population, but 29 per 100 000 in 10-17 year olds. Women are at greater risk than men, as are young subjects; the risk decreases with age. Patellar dislocation may have long-term consequences: instability of patellar origin, pain, recurrent dislocation and patellofemoral osteoarthritis.

Anatomy

Normally, the two patellar joint surfaces, medial and lateral, are symmetric and congruent with the femoral trochlea. Trochlear morphology remains relatively constant but, with growth, the thick trochlear cartilage thins in the middle, creating an illusion of a trochlear 'hollow'. The peripatellar soft tissue and particularly the medial patellofemoral ligament (MPFL) and vastus medialis obliquus (VMO) muscle contribute significantly to joint stability. The MPFL inserts to the femur between the medial epicondyle and the adductor tubercle and to the superomedial edge of the patella (Fig. 35.17).²²⁷

The MPFL provides 50-80% of the mechanisms counteracting lateral patellar glide.²²⁸ Lateral patellar dislocation renders it incompetent, promoting recurrence

of dislocation.²²⁹ Neighbouring structures, such as the patellomeniscal or patellotibial ligaments and the superficial medial retinaculum, make lesser contributions to patellar stability. The VMO muscle is very important, acting as a dynamic stabiliser, and is intimately related to the MPFL.²²⁷

Patella dislocation occurs when the patella is displaced laterally (Fig. 35.18a), leaving its confines within the trochlea groove of the femoral condyle. Acute patellar dislocation may be either traumatic with a history of a traumatic force, followed by development of a haemarthrosis or atraumatic, which usually occurs in young girls with associated ligamentous laxity. Often the latter do not have a history of significant trauma and the dislocation is accompanied by mild-to-moderate swelling.

The typical mechanism underlying patellar dislocation is a movement of the knee in flexion and valgus without direct contact, accounting for 93% of traumatic patellar dislocations.²³⁰ Most patients report a sensation of slippage, intense pain and secondary effusion, often suggestive of knee sprain. True traumatic dislocation, caused by direct tangential force dislocating the patella laterally, also occurs. Almost all patients with traumatic patellar dislocation show haemarthrosis, MPFL lesion and medial patellar wing fracture. Osteochondral fracture occurs in 25% of traumatic patellar dislocations. The recurrence risk is increased 6-fold in the case of history of ipsi- or contralateral patellar dislocation.²²⁷

Predisposing factors

Predisposing factors for patellar dislocation are either primary or secondary.²²⁷ Primary predisposing factors are:

- trochlear dysplasia
- elevated distance between the tibial tubercle (TT) and the trochlear groove (TG)
- patella alta (high patella)
- patellar tilt.

Secondary predisposing factors are:

- elevated Q angle with TT lateralisation and genu valgum
- elevated femoral anteversion with compensatory lateral tibial torsion
- vastus medialis hypoplasia
- ligament hyperlaxity with genu recurvatum
- patella dysplasia.

Clinical features

Patients with traumatic patellar dislocation usually complain that, on twisting or jumping, the knee suddenly gave way with the development of severe pain. Often the patient will describe a feeling of something 'moving' or 'popping out'. Swelling develops almost immediately. The dislocation usually reduces spontaneously with knee

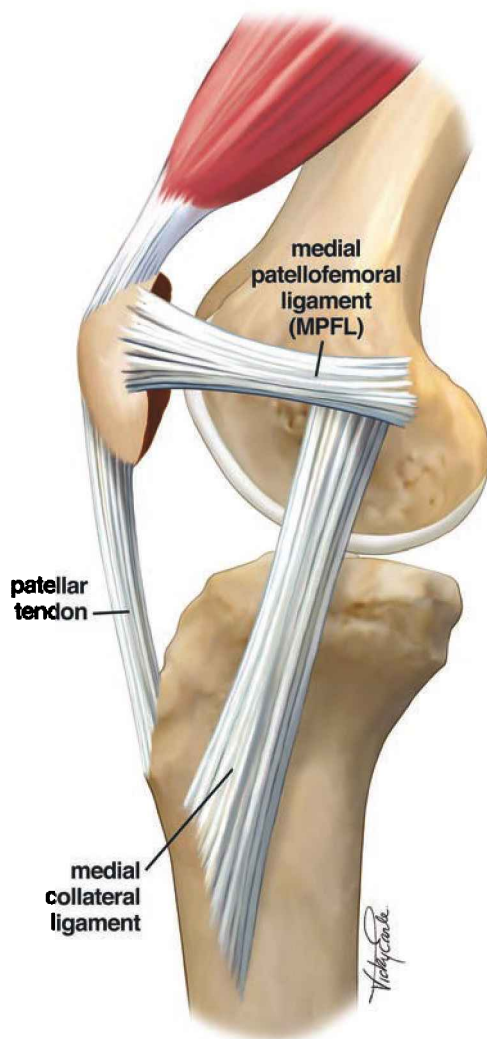


Figure 35.17 Anatomy showing the MPFL

extension; however, in some cases this may require some assistance or regional anaesthesia (e.g. femoral nerve block).

The main differential diagnosis of an acute patella dislocation is an ACL rupture. Both conditions have similar histories of twisting, an audible 'pop', a feeling of something 'going out' and subsequent development of haemarthrosis.

On examination, there is usually a gross effusion, marked tenderness over the medial border of the patella and a positive lateral apprehension test (Fig. 35.2q) when attempts are made to push the patella in a lateral direction. Any attempt to contract the quadriceps muscle

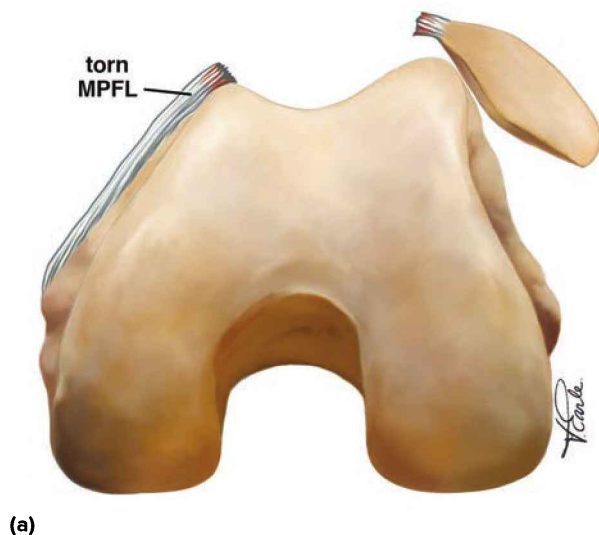
aggravates the pain and the patient often finds a painless rest in full extension.

Radiographs, including anteroposterior, lateral, skyline and intercondylar views, should be performed to rule out osteochondral fracture or a loose body.

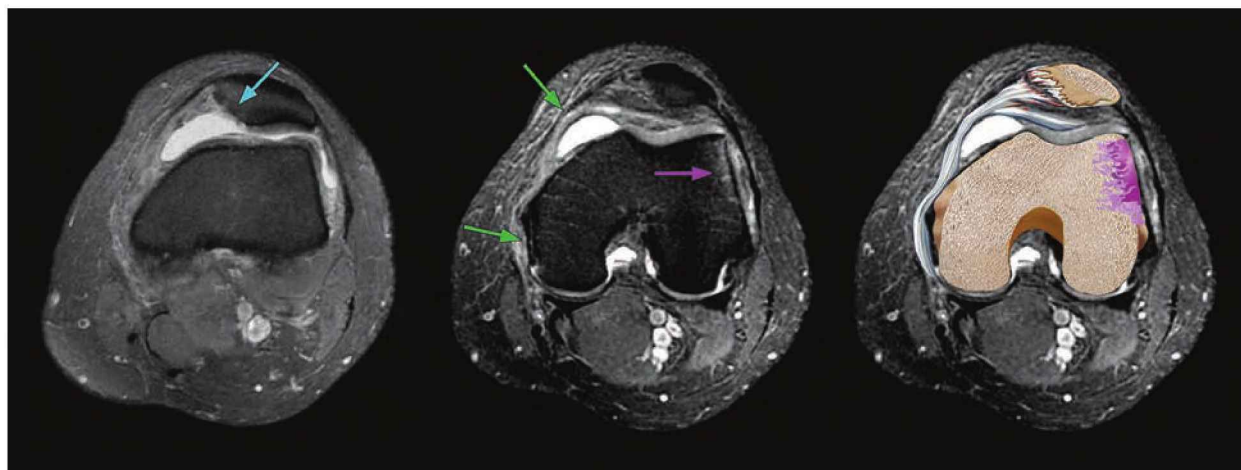
The Merchant view may reveal bone avulsion of the medial facet of the patella and enables analysis of patellofemoral centring. The avulsion involves either the patellar insertion of the MPFL or the more distal patel-lomeniscal ligament insertion. Osteochondral fractures of the inferomedial pole of the patella are highly suggestive of traumatic patellar dislocation, but are overlooked on 30–40% of initial radiographs.

Computed tomography (CT) scan can assess osseous risk factors such as patellofemoral malalignment, osteochondral defects, patellar tilt and lateral subluxation, elevated TT-TG distance and trochlear dysplasia.

MRI is more specific in precisely determining involved structures and thus guiding treatment decision making. It assesses the patellofemoral joint cartilage surfaces and also the medial patellar stabilising structures (medial retinaculum, MPFL and VMO).²³¹ In patellar dislocation, MRI reveals: haemarthrosis; osteochondral lesions and bone oedema of the medial patellar facet (Fig. 35.18b); and bone oedema of the lateral femoral condyle (Fig. 35.18c). Concave deformity of the inferomedial patella, due to impaction, is a specific sign of lateral patellar dislocation. Figure 35.18d illustrates the three components of the injury: rupture of the MPFL; injury to the inferomedial pole of the patella; and bone bruising to the femoral lateral condyle.



(a)



(b)

(c)

(d)

Figure 35.18 (a) The patella dislocates laterally tearing the MPFL (b) Axial fat-suppressed T2 MRI of the left knee. Blue arrow shows patellar chondral injury. (c) The torn MPFL is highlighted with the green arrow. Abnormal high signal represents oedema at the site of MPFL tear. Purple arrow at right of image shows bone marrow oedema at the lateral femoral condyle, the impact zone when the patella dislocates (d) Diagram demonstrating the three components of the injury—the torn MPFL, the articular damage at the inferomedial corner of the patella and the bone oedema at the lateral condyle

Treatment

The initial management of a first-time traumatic patellar dislocation is controversial, with no evidence-based consensus to guide decision making. Most first-time traumatic patellar dislocations have been traditionally treated non-operatively; however, there has been a recent trend towards initial surgical management.

Following a systematic review of 70 level I-IV studies looking at study design, mean follow-up, subjective and validated outcome measures, re-dislocation rates and long-term symptoms, the authors recommended initial non-operative management of a traumatic dislocation except in several specific circumstances.²³² These include:

- the presence of an osteochondral fracture
- substantial disruption of the medial patellar stabilisers
- a laterally subluxated patella with normal alignment of the contralateral knee
- a second dislocation
- in patients not improving with appropriate rehabilitation.

Arthroscopy should be performed in case of an intra-articular loose body secondary to a chondral lesion or osteochondral fracture. When an osteochondral fracture involves more than 10% of the patellar joint surface or involves the weight-bearing surface of the lateral femoral condyle, the osteochondral fragment should so far as possible be fixed by open surgery, for instance using resorbable pins; if this is not possible, micro-fracturing may stimulate cartilage regrowth. As the MPFL and medial retinaculum tend to be torn in acute traumatic patellar dislocation, stabilisation surgery may be required.

According to Duthon, surgery for patellar dislocation patients can be classified under three categories according to whether there are predisposing factors and whether the dislocation is inaugural or recurrent.²²⁷

1. Inaugural dislocation without predisposing factors. Immediate surgical repair of the patellar stabilising structures (VMO, MPFL, medial retinaculum) is recommended in athletic patients. Bone avulsion at the patellar insertion of the MPFL is known not to show consolidation even with proper immobilisation; functional results of non-operative treatment are unsatisfactory and fixation is recommended.
2. Inaugural dislocation with predisposing factors. Surgical repair of the patellar stabilising structures (VMO, MPFL, medial retinaculum) is recommended when risk factors for recurrence such as trochlear dysplasia, patella alta, elevated TT-TG distance, are present.
3. Recurrent dislocation. In recurrent patellar dislocation, surgery is recommended due to severe apprehension constituting a disability in everyday life, and to prevent chondral lesions which, in the long term, would cause

patellofemoral osteoarthritis. The type of surgery is determined by the presence of one or more risk factors and therefore may involve:

- lowering the tibial tubercle
- patellar tenodesis
- tibial tubercle medialisation
- sulcus-deepening trochleoplasty
- vastus medialis plasty.

While patellar dislocations are increasingly being managed with surgery, there is no good evidence of a benefit over conservative management. Hing et al. performed a literature review of five studies for the Cochrane Database.²³³ No significant difference emerged for risk of recurrent dislocation, Kujala patellofemoral score or need for surgical revision.

The most important aim of rehabilitation after patellofemoral dislocation is to reduce the chances of a recurrence of the injury. The therapist should aim to restore normal range of knee motion (full flexion and extension) and to reinforce the quadriceps to restore dynamic patellar balance.

As a result, the rehabilitation program is lengthy and emphasises core stability, pelvic positioning, vastus medialis obliquus strength and stretching of the lateral structures when tight. Similar to ACL intervention exercises, rotational control of the limb under the pelvis is critical to knee and kneecap stability. A suggested rehabilitation program is shown in Table 35.14.

Bracing could be helpful for these patients in combination with extensive rehabilitation. The patella brace usually has a hole for the patella and a lateral rim to prevent lateral dislocation. Bracing does not prevent recurrent instability but may be helpful for those with symptomatic instability.

LESS COMMON CAUSES

Patellar tendon rupture

The patellar tendon occasionally ruptures spontaneously. This is usually in association with a sudden severe eccentric contraction of the quadriceps muscle, which may occur when an athlete stumbles or when attempting a powerful take off manoeuvre (e.g. long jump event in track and field competition). There may have been a history of previous corticosteroid injection into the tendon. A previous history of patellar tendinopathy is uncommon.

Patients complain of a sudden acute onset of pain over the patellar tendon accompanied by a tearing sensation and are unable to stand. On examination, there is a visible loss of fullness at the front of the knee as the patella is retracted proximally. The knee extensor mechanism is no longer intact and knee extension with a straight leg cannot be initiated.

Table 35.14 Rehabilitation program following patellar dislocation (see Figs 35.6 and 35.7)

Phase	Goal of phase	Time post-injury	Physiotherapy treatment	Exercise program	Functional/sport-related activity
Phase 1	Control swelling Maintain knee extension Isometric quadriceps strength	0–2 weeks	Extension splint (removal dependent on surgeon/physician) Cryotherapy Electrotherapy PFJ taping Manual therapy	Quadriceps drills (supine) Bilateral calf raises Foot and ankle Hip abduction	Progress to FWB
Phase 2	No swelling Full extension Flexion to 100° 4+/5 quadriceps strength 5/5 hamstring strength	2–6 weeks	Cryotherapy Electrotherapy Compression Manual therapy Gait re-education Exercise modification	ROM drills Quadriceps/VMO setting Mini-squats and lunges Bridges (double, then single-leg) Hip abduction and extension with rubber tubing Single-leg calf raises Gait re-education drills Balance and proprioceptive drills (single-leg)	Walking Exercise bike
Phase 3	Full ROM Full strength and power Return to jogging, running, and agility Return to restricted sport-specific drills	6–8 weeks	Manual therapy Exercise/activity modification and supervision	As above—increase difficulty, repetitions and weight, where appropriate Single-leg squats Single-leg press Jump and land drills Agility drills	Straight-line jogging Swimming (light kick) Road bike Straight-line running Progressing to sport-specific running and agility (progressively sequenced) (e.g. running forwards, sideways, backwards, sprinting, jumping, hopping, changing directions, kicking)
Phase 4	Return to sport	8–12 weeks	As above	High-level sport-specific strengthening as required	Progressive return to sport (e.g. restricted training, unrestricted training, match play, competitive match play)

FWB = full weight-bearing; PFJ = patellofemoral joint; ROM = range of motion; VMO = vastus medialis obliquus

Surgical repair of the tendon is needed and must be followed by intensive rehabilitation. Full recovery takes 6–9 months and there is often some residual disability.

Bursal haematoma

Occasionally, an acute bursal haematoma or acute pre-patellar bursitis occurs as a result of a fall onto the knee. This causes bleeding into the pre-patellar bursa and subsequent inflammation. This usually settles spontaneously with firm compression bandaging. If not, the haematoma should be aspirated and the bloodstained fluid removed. Anti-inflammatory medication may also be appropriate. This injury is often associated with a skin wound (e.g. abrasion) and therefore may become infected. Adequate skin care is essential. If the bursa recurs, then aspiration followed by injection of a corticosteroid agent may be required. If conservative treatment fails, arthroscopic excision of the bursa is indicated.

Fat pad impingement

Acute fat pad impingement (often incorrectly referred to as 'Hoffa's syndrome') usually occurs as a result of a hyperextension injury. As the fat pad is the most sensitive part of the knee, this condition may be extremely painful.²³⁴ There may be an inferiorly tilted lower pole of the patella predisposing to injury. On examination, tenderness is distal to the patella but beyond the margin of the patellar tendon. A haemarthrosis may be present.

This can be an extremely difficult condition to treat. The basic principles of treatment are a reduction of aggravating activities, electrotherapeutic modalities to settle inflammation and resumption of range of movement exercises as soon as possible. Taping of the patella may help in reducing the amount of tilt and impingement (Chapter 36). If conservative management is not successful, arthroscopic joint lavage and resection of the fat pad can be helpful.

Fracture of the tibial plateau

Tibial plateau fracture is seen in high-speed injuries such as falls while skiing, wave-jumping or horseriding. This condition needs to be excluded when diagnosing collateral ligament damage with instability. The patient complains of severe pain and inability to weight-bear. Fractures are associated with a lipohaemarthrosis, which can be detected on a horizontal lateral X-ray by the presence of a fat fluid level. CT scan is helpful in defining the fracture.

Minimally displaced fractures should be treated by 6 weeks of non weight-bearing in a hinged knee brace (Fig. 35.5a). Displaced fractures (Fig 35.19) or fractures with unstable fragment/s require internal surgical fixation. Displaced vertical split fractures may be fixed percutaneously during arthroscopy.

Tibial plateau fractures are commonly associated with meniscal or ACL injuries. In these cases, arthroscopic assessment is required. Following recovery from a tibial plateau fracture, weight-bearing activity may need to be reduced as the irregular joint surface predisposes to the development of osteoarthritis.

Superior tibiofibular joint injury

Acute dislocation of the superior tibiofibular joint occurs occasionally as a result of a direct blow. The patient complains of pain in the area of the joint and may be aware of obvious deformity. The common peroneal nerve may be damaged with this injury.

Sprain of the tibiofibular joint is more common. The patient complains of local pain aggravated by movement and, on examination, there is local tenderness and some anteroposterior instability. Treatment consists of rest and local electrotherapeutic modalities. Rarely, a chronic instability of the superior tibiofibular joint develops, but these cases may need surgical stabilisation.

Ruptured hamstring tendon

Spontaneous rupture of one of the distal hamstring tendons at the knee occurs occasionally during sprinting. Sudden onset of pain is localised to either the biceps femoris tendon or the semitendinosus tendon. Pain and weakness is present with resisted hamstring contraction. When these injuries involve the tendon portion of the hamstring unit, it may be amenable to surgical exploration and repair, followed by protection in a limited-motion brace.



Figure 35.19 X-ray of a tibial plateau fracture

REFERENCES

References for this chapter can be found at www.mhhe.com/au/CSM5e